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20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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Abstract

This thesis examines how the parliamentary representation of smaller parties is affected by the transition from Hagenbach-Bischoff to Double Pukelsheim. To do so, all 31 available cantonal parliamentary elections for the eight cantons that transitioned from Hagenbach-Bischoff to Double Pukelsheim between 2007 and 2026 are analyzed. Two additional elections in Graubünden, which previously used majority voting, are presented as contextual comparisons. The smaller parties group combines all electoral lists and parties which received less than 10% of the total votes in the analyzed area. To make this comparison, the official results under Double Pukelsheim are taken and the counterfactual seat allocation under Hagenbach-Bischoff is computed. In addition are the former cantonal electoral laws included and the electoral units, observed votes, seat totals and reported list structures are held constant.

The main finding is that across all 31 elections in the main sample, the smaller parties aggregately receive a seat-share 7.6 percentage points higher under Double Pukelsheim than under the counterfactual Hagenbach-Bischoff. An aggregate gain for the smaller parties occurs in 30 of 31 elections. Further is the observed gain persistent to alternative smaller-party thresholds, across successive elections and to the omission of individual cantons. While analyses examining potential differences based on the pre-reform district magnitude or explicit electoral thresholds reveal no consistent relationship, respectively are sensitive to the inclusion of individual cantons.

The persistent mechanical seat-share gain indicates that Double Pukelsheim reduces smaller parties' representational disadvantage under Hagenbach–Bischoff. Behavioural responses to the reform remain outside the scope of the analysis.

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List of Abbreviations

AG	Canton of Aargau
AI	Artificial intelligence
ALG	Alternative – the Greens Zug (Alternative – die Grünen Zug)
BDP	Conservative Democratic Party
BGE	Decisions of the Swiss Federal Supreme Court (official collection)
CSV	Comma-separated values
CVP	Christian Democratic People's Party
DP	Double Pukelsheim system
EQ1	Exploratory question 1 (district magnitude)
EQ2	Exploratory question 2 (explicit electoral threshold)
EQ3	Exploratory question 3 (successive elections)
FDP	FDP.The Liberals
FSC	Swiss Federal Supreme Court
GLP	Green Liberal Party
GR	Canton of Graubünden
HB	Hagenbach-Bischoff method
HSG	University of St. Gallen
JFSH	Young Liberals Schaffhausen (Jungfreisinnige Schaffhausen)
JUSO	Young Socialists Switzerland
LLM	Large language model
NA	Not available or not applicable (no reported value in the appendix tables)
NW	Canton of Nidwalden
PDF	Portable Document Format
pp	percentage points
PR	proportional representation
Q1	First quartile (25th percentile)
Q3	Third quartile (75th percentile)
SD	Swiss Democrats (Schweizer Demokraten)
SH	Canton of Schaffhausen
SP	Social Democratic Party of Switzerland
SVP	Swiss People's Party
SZ	Canton of Schwyz
UR	Canton of Uri
VS	Canton of Valais
XLSX	Excel workbook file format (Office Open XML)
ZG	Canton of Zug
ZH	Canton of Zurich

1 Introduction

On the 18th December of 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 I 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds in some districts were up to 34% high. Furthermore varied the thresholds heavily across the districts. Consequently, votes did not have equal effect, which undermined the electoral equality included in Art. 8 (Equality before the law) and Art. 34 (political rights) of the Federal Constitution. Based on this criticism the court decided that the City of Zurich had to implement corrective measures, but let the city choose the way of reform (BGE 129 I 185 E. 7.3, E. 7.6.2–7.6.3 and E. 8.3.2; Schweizerische Eidgenossenschaft, 1999, Arts. 8 and 34). The Government of the Canton of Zurich decided to keep the constituency structure and instead commissioned the German mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a method called "biproportional divisor method with standard rounding" which is today often simply called "Double Pukelsheim" or "Biproportional apportionment". (Regierungsrat des Kantons Zürich, 2014, pp. 3–4; Lüscher, 2024, pp. 9-10)

The distinguishing feature of the Pukelsheim Proporz compared with Hagenbach-Bischoff is the so called upper-apportionment. It artificially creates a single electoral district for the initial allocation of the parliamentary seats to the parties or list groups. Through this mechanism, the difference between seat and vote shares gets minimized and with that also the effective threshold for parties to enter the parliament. (Lüscher, 2023, p. 575) This mechanism and the lowered threshold make the reform especially relevant for the representation of smaller parties, since votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level. Sandro Lüscher illustrates this mechanical effect with the case of the Schwyz Kantonsrat election in 2016. In that election, the Green Liberal Party received three seats, whereas under the old HB system it would have received none (Lüscher, 2025, p. 1-3).

Since Zurich first used the system in 2007, eight further cantons have applied it: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016), Valais (2017), Uri (2020) and Graubünden (2022). In 2025 also the voters from Basel-Landschaft decided to implement the biproportional apportionment from 2027 on (Lüscher, 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b). While nine of the swiss cantons have already implemented the DP system, in Thurgau, Lucerne and St.Gallen, the application is currently under debate (Grosser Rat des Kantons Thurgau, n.d.; Regierungsrat des Kantons Luzern, 2026, pp. 19–20 ; Initiativkomitee «Damit jede Stimme zählt», n.d.).

The number of cantons who have already implemented the system or are debating about doing so suggests that the decision by the FSC had a substantial impact on how Swiss citizens elect their regional parliaments today. It seems that the Double Pukelsheim through its wide implementation across cantons is a success story. Since the first election under Double Pukelsheim in Zurich in 2007, the

[1]The 33 elections comprise Zurich (2007, 2011, 2015, 2019, and 2023; Amt für Statistik und Daten Kanton Zürich, 2026); Schaffhausen (2008, 2012, 2016, 2020, and 2024; Kanton Schaffhausen, n.d.; Staatskanzlei Kanton Schaffhausen, 2020, 2024); Aargau (2009, 2012, 2016, 2020, and 2024; Departement für Finanzen und Ressourcen des Kantons Aargau, 2009; Regierungsrat des Kantons Aargau, 2013; Statistisches Amt Kanton Aargau, n.d.); Nidwalden (2014, 2018, 2022, and 2026; Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026); Zug (2014, 2018, 2022, and 2026; Kanton Zug, 2022a, 2022b, 2022c,

system has been used in 33 cantonal parliamentary elections.[1] Of these, 31 elections in eight cantons form the main sample.

Taken together, this fairly big number of elections and almost 20 years of experience provide a reason to evaluate the impact of the reform, especially on smaller parties. How much has the reform changed smaller' parties seat shares and how large is the actual difference in seat allocation between the Hagenbach-Bischoff and Double Pukelsheim systems? Are there other major differences, for example based on the pre-reform district sizes, and how do smaller parties' seat shares evolve across successive elections?

Existing research primarily examines Double Pukelsheim in legal, normative, or system-level terms. A few studies also conduct counterfactual comparisons, but these cover either individual elections or only one election per canton (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592; Leuzinger, 2018, p. 42; Lüscher, 2024, p. 13). Although their results indicate that smaller parties can benefit from Double Pukelsheim, the magnitude and consistency of this mechanical allocation difference across successive elections remain less systematically examined.

This thesis addresses this gap by applying a consistent Hagenbach–Bischoff counterfactual to the 31 available elections in the eight cantons that transitioned from HB to DP. Two Graubünden elections are presented as contextual comparisons but excluded from the main transition estimate. Holding the observed post-reform votes and list structure constant, the analysis measures the difference in the share of seats won by smaller electoral lists and examines variation across institutional settings and successive elections.

2 Theory and Literature Review

2.1 Institutional and Theoretical Background

This chapter provides the institutional and theoretical background for analyzing the transition from Hagenbach-Bischoff to Double Pukelsheim and reviews the relevant literature. It first outlines the historical development of electoral systems in Switzerland and then explains the functioning and limitations of the Hagenbach-Bischoff and Double Pukelsheim methods. It subsequently examines the adoption and canton-specific implementation of Double Pukelsheim. Finally, the literature review situates this thesis within previous research before the expected effects of the transition are derived.

2.1.1 Historical Development

Until the late nineteenth century Swiss voters elected their regional parliaments only under majority voting. In a majority voting system, the candidates or parties who receive the relative or absolute majority receive the mandates, which leads to a systematic favouring of the dominant parties. Over time, this inequality led to dissatisfaction among smaller parties. Subsequently, these demanded a change to a proportional electoral system. The calls for change led; to that most of the cantons' Grand Councils as well as the National Council were later elected with the Hagenbach-Bischoff or another proportional system. In the beginning of the 21st century, the cantons of Graubünden and Appenzell Innerrhoden were the only ones to still elect their parliaments with the majority system. (Gilg, 2014, "Mehrheits-

2026); Schwyz (2016, 2020, and 2024; Staatskanzlei Kanton Schwyz, 2016, 2020, 2024); Valais (2017, 2021, and 2025; Kanton Wallis, 2016, 2020, 2024; Staat Wallis, 2017); Uri (2020 and 2024; Standeskanzlei Uri, 2024a, 2024b); and Graubünden (2022 and 2026; Kanton Graubünden, 2022, 2026).

und Verhältniswahl" section ; Bundeskanzlei, 2013, p. 2) Nowadays the canton Appenzell Innerrhoden is the only remaining canton voting the Grand Council by majority voting (Bundeskanzlei, 2025, p. 18). It can therefore be said that proportional systems have become the dominating electoral systems for cantonal parliaments in Switzerland.

In the last 20 years the prior dominance of the Hagenbach-Bischoff system has been substantially reduced by the wide implementation of the Pukelsheim Algorithm. As of 2025, eleven cantons apply the parliamentary seats with Hagenbach-Bischoff and ten cantons with the DP; Glarus and Basel-Stadt use the Sainte-Laguë method, Ticino and Vaud use the Hare/Niemeyer method and Appenzell Innerrhoden majority voting (Bundeskanzlei, 2025, p.18). Both Hare/Niemeyer and Sainte-Laguë are generally associated with a higher proportionality than the HB system; therefore this might partly explain why none of the cantons using these two PR-Systems have adopted the Double Pukelsheim system (Bundeskanzlei, 2025, pp. 22; Deutscher Bundestag, 2007, p.9). This means that of the nine cantons covered in the thesis, the eight from the main sample previously used predominantly Hagenbach-Bischoff. The special cases are Graubünden, Zug, Uri and Schaffhausen where majority voting had been used either canton-wide or for the three latter ones in at least one constituency. (Bundeskanzlei, 2013, Table 1; BGE 136 I 376 E. 5.1 ; Standeskanzlei Uri, 2020, p. 1; Statistisches Amt des Kantons Zürich, 2011, p. 286; Regierungsrat des Kantons Schaffhausen, 2007, p. 6; Regierungsrat des Kantons Aargau, 2007, p. 18).

2.1.2 The Hagenbach-Bischoff Method

To be able to comprehend later the differences between the HB and the DP Method, one has to understand the allocation under the priorly used Hagenbach-Bischoff system.

Under Hagenbach-Bischoff, seats are allocated separately in each constituency. Let M denote the number of mandates in the constituency, v_i the votes for an eligible list i, V the total number of valid list votes of all lists in the constituency used to calculate the distribution number and D the distribution number for the smallest integer strictly greater than $V/(M+1)$. In the first allocation, each list receives the integer part of its votes divided by D. (Bundeskanzlei, 2025, p. 41)

$$D = \left\lfloor \frac{V}{M+1} \right\rfloor + 1, \qquad s_i = \left\lfloor \frac{v_i}{D} \right\rfloor.$$

If the first allocation leaves seats unassigned, the remaining seats are awarded one at a time. For each list, its votes are divided by the number of seats it currently holds plus one. The list with the highest quotient receives the next seat; its quotient h_i and the allocated seats s_i are then recalculated, and the process continues until all M seats have been filled:

$$h_i = \frac{v_i}{s_i + 1}.$$

It should be noted that the HB system calculation produces the same allocation as the D'Hondt divisor method with downward rounding (Bundeskanzlei, 2025, p. 41). Furthermore there are factors like list alliances, explicit electoral thresholds, and rules for resolving equal quotients which depend on the electoral law of the corresponding canton. The counterfactual in this thesis applies the relevant former cantonal rules within each constituency, as detailed in Section 4.2 .

The central weakness of the traditional cantonal application of Hagenbach–Bischoff arises from applying the formula separately across constituencies which often have few seats. Small district magnitudes create high natural thresholds, meaning that votes for smaller parties may fail to contribute to a seat. This might hinder equality of electoral success and systematically disadvantage smaller parties. As shown in the later section on the adoption of Double Pukelsheim, these inequalities were central to the legal and political debates that led several cantons to adopt Double Pukelsheim. (Pukelsheim & Schuhmacher, 2004, pp. 507-508, 521; BGE 129 I 185 E. 7.1.1 and E. 7.6.2)

2.1.3 The Double Pukelsheim Method

The following section explores the mechanism of the Double Pukelsheim and its differences to the Hagenbach-Bischoff method.

Based on Articles 8 and 34 of the Federal Constitution, there are three dimensions of electoral equality, which have to be followed. These are the principle of equal voting value, equality of voting power and equality of electoral success. The equal voting value ensures that each voter within an electoral district gets allocated the identical number of votes. The second principle requires the population represented per seat to be approximately equal across electoral districts. Thirdly the equality of electoral success should assure that all votes contribute equally to the election and that as many votes as possible are included in the allocation of the mandates. (Pukelsheim & Schuhmacher, 2004, pp. 506-507) Districts with fewer than nine seats have a natural thresholds above 10%, making it difficult for smaller parties to enter the seat allocation in that district. The FSC treats 10% as a target for natural thresholds when redesigning an electoral system. (Pukelsheim & Schuhmacher, 2004, p. 505; BGE 131 I 74 E. 5.2–5.4) For the traditional application of Hagenbach–Bischoff, the main difficulty is that small districts contradict the 10% target. (Pukelsheim & Schuhmacher, 2004, p. 508; BGE 129 I 185 E. 7.1.1 and E. 7.6.2)

To approach this issue, the Double Pukelsheim uses an upper and a lower apportionment (Pukelsheim & Schuhmacher, 2004, p.513).

- **Upper Apportionment** To bypass the natural thresholds, in the upper apportionment, the whole canton is seen as one single district. Seats are allocated to parties in proportion to their cantonal vote shares. To achieve this, the votes from the district-level have to be summarized on a cantonal level. Because of the different district magnitudes, voters receive a different amount of votes. Therefore the voter equivalents have to be obtained, which is done as followingly:

$$w_{ij} = R_c\left(\frac{v_{ij}}{M_j}\right), \qquad W_i = \sum_j w_{ij}.$$

  Here, i denotes a party, j a constituency, v_{ij} a party's votes in a specific district and $M_j$is the number of mandates in the constituency. R_c represents the applicable cantonal rule: leaving the qoutient unrounded or rounding to the nearest integer, and W_i is the party's aggregated voter equivalent. (Pukelsheim & Schuhmacher, 2011, pp. 1584–1585, 1594–1595) Each party's total seat entitlement is then calculated followingly:

$$S_i = \text{round}\left(\frac{W_i}{d}\right)$$

  The divisor d needs to be set so that exactly as many seats can be allocated as there are available. (Pukelsheim & Schuhmacher, 2011, p. 1585)

- **Lower Apportionment** The lower apportionment assigns each party's cantonwide seats from the upper apportionment to its district lists while preserving each district's predetermined seat total. For that first the party votes in each district are divided by a district divisor which ensures that the exact amount of seats are allocated. The results are then rounded. (Pukelsheim & Schuhmacher, 2011, p. 1585). However, the resulting party totals may differ from the upper apportionment results. To correct such differences, each party's vote total in every district is divided by the same additional list group divisor. (Pukelsheim & Schuhmacher, 2011, p. 1586) With a_j denoting the district divisor, b_i the list-group divisor, party i receives the following seats in district j:

$$s_{ij} = \text{round}\left(\frac{v_{ij}}{a_j b_i}\right).$$

(Pukelsheim & Schuhmacher, 2004, p. 515)

An hypothetical example illustrates this correction. A party has received in the lower apportionment one seat more than in the seat entitlement from the upper apportionment. Let's say this party has in District Y a quotient after division by the district divisor of 2.51. With the standard rounding, the party would receive three seats in this district. Given that the party's quotient is closest to downrounding in that district, one would choose a number which rounds the quotient down to two seats, e.g. with $2.51/1.01 \approx 2.49$. However, this change might cause that the district gets allocated one seat too few. Both sets of divisors must therefore be adjusted until the party entitlements S_i and district seat totals M_j are simultaneously satisfied (Pukelsheim & Schuhmacher, 2011, p. 1586):

$$\sum_j s_{ij} = S_i \quad \text{for every} \quad i, \qquad \sum_i s_{ij} = M_j \quad \text{for every} \quad j.$$

(Pukelsheim & Schuhmacher, 2004, p. 515)

To summarize, the upper apportionment in the Double Pukelsheim system reduces inequalities in electoral success caused by the high natural thresholds of small constituencies under the previous district-by-district Hagenbach-Bischoff system. It does so by pooling parties' votes across the canton before seats are assigned to constituencies. This is also the main improvement of DP regarding the three dimensions of electoral equality in contrast to Hagenbach-Bischoff.

2.1.3.1 Trade-offs and Criticisms

Like every proportional system, the Double Pukelsheim system has flaws for which it gets criticized. One of the main weaknesses of the biproportional apportionment compared to the HB System is its complexity. It gets mentioned throughout that the system is relatively difficult to understand and compute. (Bundeskanzlei, 2013, p. 19 ; Bundeskanzlei, 2025, p.24 ; Lüscher, 2024, p.20) As a counterargument, Lüscher mentions that the list alliances used under Hagenbach-Bischoff also led to a high complexity situation. This especially since before an election, it was impossible to estimate, which party would profit from the alliance. (Lüscher, 2024, p.20) Another main point of criticism is the so called "discordant seat allocations". They describe the phenomenon when within a district a party gets allocated more seats than another party with more votes in that district. This phenomenon can appear through the lower apportionment process and the dual role of the system to achieve proportionality both on the cantonal and district-level. The emerging situation can cause confusion and frustration especially within the party which received more votes but less seats in that specific

district. (Pukelsheim & Schuhmacher, 2004, p. 519) Also for the voters can this situation be confusing, especially since in Switzerland elections often contain a personal vote component. A reaction to this flaw of the DP-System was that some cantons implemented a majority clause. The clause aims to ensure that the strongest list in the corresponding constituency receives at least one seat. (Lüscher, 2024, p. 21) The option of implementing a majority clause to solve this issue was also pointed out by Friedrich Pukelsheim himself (Pukelsheim & Schuhmacher, 2004, p. 520). Another issue of the DP system is that different turnout levels and different proportions of residents who are eligible to vote in the constituencies can determine the allocation of mandates among parties in other constituencies. To illustrate this, Lüscher provides an example: There are three districts with equal population and number of mandates. District 1 produces 50% of all votes, while District 2 and 3 only produce each 25%. Because of the higher voter turnout, District 1 contributes 50% of the votes used to determine the canton-wide allocation of seats among parties, while only having 33.3% of the mandates. This has the effect that if a party was more successful in District 1 than in the others, it will be overrepresented in the other districts. (Lüscher, 2023, pp. 578-579) One other point which was often discussed to be a possible weakness of the Double Pukelsheim is a possible Party System Fragmentation. For example both reports of the Swiss Federal Chancellery in 2013 and 2025 mention it and introduce the usage of explicit thresholds to prevent it. (Bundeskanzlei, 2013, p.18 ; Bundeskanzlei, 2025, p. 24) Despite these concerns, Sandro Lüscher came to the perception that the general trend of a rising amount of parties in Swiss cantonal parliaments is unrelated to the implementation of the Double Pukelsheim. He then also mentioned that the inexistency of an effect might come from that several cantons have implemented thresholds. (Lüscher, 2024, p. 135) Therefore it can not be conclusively stated whether the reform would have had an impact on party system fragmentation without the implementation of thresholds or not.

Taken together, these criticisms illustrate the central trade-offs of Double Pukelsheim. Although the system improves canton-wide proportionality and reduces the number of ineffective votes, it does so at the cost of greater complexity and potentially counterintuitive outcomes at the constituency level. Electoral thresholds can mitigate discordant seat allocations and party-system fragmentation, but may simultaneously weaken proportionality. The system therefore does not eliminate electoral distortions entirely.

2.1.4 The Adoption of Double Pukelsheim in Swiss Cantons

As outlined in the section Historical Development, did the Double Pukelsheim gain popularity after its initial implementation in Zurich. The form, reason, and circumstances under which its applied, though, vary from canton to canton. The subsequent table provides an overview of the different ways of implementation across the cantons.

Table 1: Adoption of Double Pukelsheim and explicit electoral thresholds by canton.

Canton	First election under DP	Main reform driver	Explicit electoral thresholds
Zurich	2007	Reaction to the FSC ruling for the City of Zurich, since the canton faced similar issues	5% district-level (2007-2019), 5% district-level or 3% cantonal-level (2023-)
Schaffhausen	2008	Parliament was downsized; subsequently reforms were needed; In view of the FSC judgements in Zurich and Aargau, Double Pukelsheim was chosen	None
Aargau	2009	Parliament was downsized, no Action regarding electoral equality was taken, the FSC ruled for change and followingly the implementation of DP was decided	None (2009), 5% district-level or 3% cantonal-level (2012-)
Nidwalden	2014	Voters appeal at cantonal court and later at FSC; FSC decided in favour of the appeal; the DP system was favoured as a solution and implemented	None
Zug	2014	Voters and parties appeal at the FSC; Court agrees on the main point; the DP system is favoured and implemented	5% district-level or 3% cantonal-level
Schwyz	2016	Voters appeal at the FSC based on the new constitution and its districts; FSC agreed partly; multiple options were explored; finally the DP-system was implemented.	1% cantonal-level
Valais	2017	FSC appeal in 2004 was rejected; a second appeal was approved by the FSC; subsequently implementation of DP was done via a temporary decree because of lacking electorate approval; later the system was officially incorporated	8% sub-district-level
Uri	2020	Appeal at the FSC was partially upheld; subsequently Double Pukelsheim was implemented for municipalities with at least five seats	None
Graubünden	2022	Voters and parties appeal at FSC was partially upheld; multiple options were evaluated; finally transition from Majority to DP system was decided	3% cantonal-level
Basel-Landschaft	2027	A single parliamentarian advocated for the Double Pukelsheim over multiple years; subsequently the implementation was proposed and approved by Parliament and voters	5% district-level or 3% cantonal-level

[2]

[2] Zurich: (BGE 129 I 185 E. 7.1.2, E. 7.6.2–7.6.3 and E. 8.3.2; Lüscher, 2024, pp. 9–10, 33-36 ; Pukelsheim & Schuhmacher, 2004, p.505 ; Regierungsrat des Kantons Zürich, 2024, pp. 2,5) ; Schaffhausen: (Regierungsrat des Kantons Schaffhausen, 2007, pp. 1, 3, 13-14; Lüscher, 2024, pp. 41- 43), Aargau: (Lüscher, 2024, pp. 37-40); Nidwalden: (BGE 136 I 352 E. 4.; Lüscher, 2024, pp. 44–45), Zug: (BGE 136 I 376 E. 4.5; Lüscher, 2024, p. 46-51) ; Schwyz : (Lüscher, 2024, pp. 52-55 ; Urteil des Bundesgerichts 1C_407/2011 etc. vom 19. März 2012, Sachverhalt A–B und E. 5.3, 6) ; Valais : (BGE 131 I 85 E. 2.3–2.5; Lüscher, 2024, pp.57-63 ; Kanton Wallis, 2004b, Art. 154 Abs. 1) ; Uri: (BGE 143 I 92 Sachverhalt, E. 5–5.4 and E. 6.5 ; Kanton Uri, 2019, pp. 4-7 and 10-11 ; Lüscher, 2024, pp. 63–67) ; Graubünden: (BGE 145 I 259 E. 7.3 and E. 8.5.3; Kanton Graubünden, 2021/2026, Art. 26 para.1 ; Lüscher, 2024, pp. 68 and 71-72) ; Basel-Landschaft: (Lüscher, 2024, p.86;

To summarize, in eight of the ten cantons which adopted DP, a raised complaint at the FSC led to a later reform. Only in the cantons of Schaffhausen and Basel-Landschaft did the reform process take place without a Federal Supreme Court ruling. Nevertheless the canton of Schaffhausen's decision was directly influenced by the FSC's decisions in Aargau and Zurich. Throughout all complaints, the concern was electoral equality. In seven of them excessively high natural thresholds in proportional constituencies were central. In Graubünden, though, the concerns were mainly raised by its use of the majority system for its large districts and the allocation of seats among constituencies. This demonstrates the influence of the 2002 Federal Supreme Court decision regarding Zurich's City Parliament.

Beyond their different paths to reform, the cantons also implemented Double Pukelsheim under substantially different institutional conditions. The number of seats included in the Double Pukelsheim allocation varies from 37 in Uri, where the system applies only to four municipalities, to 180 in Zurich (Standeskanzlei Uri, 2020, p. 1; Kanton Zürich, 2003/2021, § 88). Moreover, some cantons introduced explicit local or canton-wide electoral thresholds, whereas others did not, potentially limiting smaller parties' ability to benefit from the more proportional allocation method. These arrangements have also changed over time for example through the later introduction of thresholds or majority clauses while party competition and vote distributions vary between elections. This institutional variation motivates the later exploratory comparisons : Whether smaller parties' seat-share gains differ by the presence of an explicit electoral threshold and across successive elections within the same canton.

2.2 Literature Review

Building on the institutional mechanisms described above, the existing empirical literature generally finds that Double Pukelsheim improves vote–seat proportionality and benefits parties that were disadvantaged by small electoral districts. However, the studies differ in their research designs, geographical coverage, and treatment of counterfactual electoral rules.

Pukelsheim and Schuhmacher provide an early counterfactual comparison based on the 2011 Zurich election. They apply the former Hagenbach–Bischoff method to the observed votes while assuming four hypothetical list alliances based on political orientations and historical alliances. Under this counterfactual, the larger SVP, SP, and FDP receive additional seats, primarily at the expense of smaller parties such as the BDP and Alternative List. Their calculation also indicates considerably greater inequality in the electoral success of votes under Hagenbach–Bischoff. The analysis nevertheless concerns only one election, and its results depend partly on assumptions about which list alliances parties would have formed. (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592)

Lüscher conducts a similar comparison for the 2016 Schwyz election. Under his Hagenbach–Bischoff counterfactual, which does not assume any list alliances, the two biggest parties, the SVP and CVP, would have benefited, whereas the left alliance and the GLP were disadvantaged. Especially extreme is the difference between the three GLP seats under Double Pukelsheim versus zero seats under the counterfactual HB system. (Lüscher, 2024, p. 13) This example illustrates the potentially substantial effect of the allocation rule on an individual smaller party, but it does not establish whether effects of this magnitude occur systematically across cantons and elections.

Regierungsrat des Kantons Basel-Landschaft, 2024 ; Landrat des Kantons Basel-Landschaft, 2024, p. 1052 ; Kanton Basel-Landschaft, 2025b ; Kanton Basel-Landschaft, 1981/2025, § 39a para. 1(a–b) ; Kanton Basel-Landschaft, 2026)

Leuzinger broadens the comparison by examining the most recent election in each of the seven cantons that had introduced Double Pukelsheim by 2018. His study is the closest precedent for the present analysis because it calculates hypothetical seat distributions for multiple elections under the former cantonal electoral systems. However, its analysis is limited to seven elections and it assumes that the list alliances observed in the final pre-reform election would have persisted. (Leuzinger, 2018a, p. 42) The present thesis does not carry these historical alliances forward. Consequently, the two counterfactual specifications can produce different Hagenbach–Bischoff seat allocations. As a result, there were differences in the distribution of seats between Leuzinger's study and this study for the cantons that previously used list alliances (Leuzinger, 2018b). This illustrates the sensitivity of counterfactual results to assumptions about unobserved alliance formation. Leuzinger also mentions that he observed that mainly smaller parties profited from the transition and that around 10% of the mandates mechanically change the party through the transition (Leuzinger, 2018a, p.42).

Taken together, the counterfactual calculations by Pukelsheim & Schuhmacher, Lüscher, and Leuzinger constitute the closest methodological precedents for the present thesis. They apply a former allocation rule to votes observed under Double Pukelsheim and compare the resulting counterfactual seat distribution with the actual allocation. Their comparisons therefore hold the observed post-reform vote distribution constant and only isolate the difference produced by the allocation rules.

The interpretation of this difference can be clarified using Fiva and Folke's framework for separating the mechanical and psychological effects of electoral reform. Fiva and Folke examined a reform of Norwegian municipal elections that replaced D'Hondt with modified Sainte-Laguë. (Fiva & Folke, 2016, p.266) Their principal estimate of the mechanical reform effect compares the actual pre-reform allocation with a counterfactual application of the new formula to the same pre-reform votes. They also apply the former and new formulas to the same post-reform votes to examine how behavioral responses may alter the magnitude of the mechanical effect. (Fiva & Folke, 2016, pp. 270–272) The Swiss studies described above, and the present thesis, follow this latter computational approach. There the former formula is applied counterfactually to votes observed under the new system. This measures the mechanical allocation difference conditional on the observed post-reform vote distribution.

Rather than recalculating individual elections, Lüscher's principal empirical analysis estimates how cantonal political outcomes would have developed without the reforms. He finds that Double Pukelsheim substantially reduced vote–seat disproportionality, with stronger improvements in cantons that previously had smaller average district magnitudes (Lüscher, 2024, pp. 129–143). However, Lüscher finds no significant increase in the effective number of parliamentary parties or electoral candidacies. Voter turnout initially declined slightly, but this effect diminished over subsequent elections. These findings distinguish the reform's immediate mechanical consequences from its more limited behavioural effects on parties and voters.(Lüscher, 2024, pp. 143–145)

The existing literature therefore provides strong evidence that Double Pukelsheim improves proportionality, as well as multiple examples in which smaller parties gain seats. What remains less systematically examined is the magnitude and consistency of the conditional mechanical allocation difference across successive elections. Existing counterfactual studies either focus on only a small number of individual elections or focus on different aspects like Lüscher.

This thesis addresses that gap by applying a consistent Hagenbach–Bischoff counterfactual to all 31 available elections in the eight cantons that transitioned from HB. Two elections from Graubünden are additionally presented as hypothetical contextual comparisons, since the canton priorly used majority voting. The thesis holds observed post-reform votes constant, applies a common definition of smaller parties, and measures their change in seat share for every election. In contrast to Leuzinger (2018a) and Pukelsheim & Schuhmacher (2011), it does not assume that hypothetical or historical pre-reform alliances would have persisted. The analysis therefore estimates the mechanical allocation difference conditional on the observed votes without introducing an additional behavioral assumption about counterfactual list-alliance formation.

2.3 Expected Effects of the Transition

Based on the theoretical considerations outlined in the preceding sections, several effects can be expected from the transition of the Hagenbach–Bischoff method to the Double Pukelsheim system. Under a district-by-district Hagenbach–Bischoff allocation, small district magnitudes create high natural thresholds for winning a parliamentary seat. Votes for parties which fail to win a seat in a district will not contribute elsewhere. This produces ineffective votes and therefore disproportionately disadvantages smaller parties. The biproportional apportionment instead through the upper apportionment pools votes across the whole canton, before it distributes each party's seats among the districts. This should increase the equality of electoral success and reduce vote–seat disproportionality. The changed seat-allocation mechanism provides the basis for the main hypothesis. When holding the observed votes constant, smaller parties are expected to aggregately receive a higher seat-share under Double Pukelsheim than under the counterfactual Hagenbach-Bischoff across the analyzed elections. (Pukelsheim & Schuhmacher, 2004, pp. 506–508, 513-514 , 518)

Followingly to the prior explanation, the changed seat allocation procedure should have a higher effect in cantons with many low-magnitude districts. This expectation is consistent with Lüscher's finding that the reduction in disproportionality was stronger in cantons with smaller pre-reform average district magnitudes (Lüscher, 2024, pp. 132–133). Furthermore, the effect is expected to depend on the specific institutional design adopted by each canton. Reflecting Switzerland's federal structure, the cantons that have adopted DP differ in their use of canton-wide and district-level legal thresholds, majority clauses, and other supplementary adjustments. Explicit electoral thresholds can restrict gains among smaller parties by excluding lists that do not reach the required vote share and may therefore partly offset the greater proportionality of the DP system. (Bundeskanzlei, 2013, p. 18; Lüscher, 2024, pp. 107, 135) In the Zurich 2011 election, for example, the SD and Piraten would have received a seat if there had not been a 5% district-level threshold (Pukelsheim & Schuhmacher, 2011, p.1593). The majority clause in constrast has no direct effect on the smaller party seat-share gain, since the cantons ensure in their law that the seat entitlements according to the upper apportionment must hold. Zug's law for e.g. explicitly states that the majority clause gets deactivated if it is essential to fulfill the upper apportionment seat entitlement. (Kanton Schaffhausen, 1904/2019, Art. 2d para. 1bis and 2(b); Kanton Zug, 2006/2026, § 52e para. 1a, 1b and 2(b); Kanton Graubünden, 2021/2026, Art. 28 para. 2 and 4(b))

Valais is a special case, since it pools the votes of 14 subdistricts across 6 electoral districts and not across the whole canton (Lüscher, 2024, p.57). This is expected to limit the proportionality-enhancing effect relative to a canton-wide implementation, especially for parties whose support is dispersed across regions.

Comparisons across successive elections also provide exploratory context by examining whether smaller parties' mechanical seat-share gains persist or vary over time. Since the vote distribution and the composition of the smaller-party group vary and electoral rules can change, no directional expectation is formed.

The estimated difference may additionally depend on the electoral rules used to construct the counterfactual outcomes, including the treatment of list alliances and canton-specific allocation provisions. Together, these expectations provide the foundation for the following hypothesis and exploratory questions concerning differences in smaller parties seat shares.

3 Main Hypothesis and Exploratory Questions

To answer the research question, the thesis tests one main hypothesis concerning how the transition from the Hagenbach–Bischoff system to Double Pukelsheim mechanically affects the seat shares of smaller parties. Complementary exploratory analyses evaluate whether this effect varies with district magnitude, the presence of an explicit electoral threshold, and successive elections since implementation within each canton. Together, these analyses identify both the immediate mechanical and potentially evolving consequences of the electoral-system reform. For all analyses, Graubünden is shown as a contextual comparison and not included in the computational summaries.

3.1 Main Hypothesis

The cantonwide pooling under the Double Pukelsheim system is expected to improve smaller parties' collective representation. Based on this expectation, the main hypothesis examines whether smaller parties, when holding votes constant, collectively receive a larger seat share under Double Pukelsheim than under the counterfactual Hagenbach-Bischoff allocation.

In order to be able to compare this, in each election e, smaller parties or electoral lists are those with less than 10% of the vote in the *analyzed area*. The variable *analyzed area* describes the whole canton except in Uri, where the *analyzed area* includes only the four municipalities using the DP. The 10% cutoff is chosen with reference to the Swiss Federal Supreme Court's benchmark that natural (implicit) thresholds should not exceed 10% when an electoral system is redesigned (BGE 131 I 74 E. 5.3–5.4; Bundeskanzlei, 2013, p. 8). This benchmark motivates the definition of smaller parties used in this analysis.

Let S_e^{DP} and S_e^{HB} denote the aggregated number of seats allocated to smaller parties in election e under Double Pukelsheim and the counterfactual Hagenbach–Bischoff system, respectively. Also let M_e denote the total number of seats included in the comparison. Smaller parties' combined gain in seat share, shown for each election in Figure 1, is calculated as follows:

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points} .$$

The hypothesis uses the unweighted mean, $\bar{\Delta} = \frac{1}{E}\sum_{e=1}^{E}\Delta_e$, so each election counts equally. The analysis also reports a mean weighted by the number of analyzed seats in each election for comparison.

This leads to the following directional hypothesis:

Main Hypothesis: Across the analyzed elections, the mean mechanical difference $\bar{\Delta}$ is positive: electoral lists with less than 10% of the vote aggregately receive a larger share of

the analyzed seats under Double Pukelsheim than under the counterfactual Hagenbach–Bischoff allocation.

3.2 Exploratory Questions

3.2.1 Exploratory question 1: Differences by district magnitude

Smaller districts are connected to higher natural thresholds; therefore its expected that the advantage from pooling votes on a cantonal level might be larger in cantons which previously had smaller districts. To examine this, the analysis looks at how the smaller parties average seat-share gains vary with the pre-reform district structure.

The analysis is done with two measures; for both the average smaller-party gains are compared between cantons below and those at or above the mean across the eight cantons. The primary measure is the mean baseline district magnitude and the secondary measure is the relative share of districts with less than 9 seats. The secondary measure adds context and can reveal otherwise unobserved patterns for example caused by outliers. Both district measures use a fixed historical baseline from each canton's last pre-reform election. For canton c, with $D_{c,0}$ baseline districts and district magnitudes $m_{dc,0}$, the mean district magnitude is

$$\bar{m}_{c,0} = \frac{1}{D_{c,0}} \sum_{d=1}^{D_{c,0}} m_{dc,0} = \frac{M_{c,0}}{D_{c,0}}.$$

Directional Expectation: A lower baseline mean district magnitude is expected to be associated with a larger average gain for smaller parties.

The secondary measure is defined followingly: c denotes the canton and 0 its pre-reform baseline; $D_{c,0}$ is the number of districts, and $m_{dc,0}$ their seat magnitudes.

$$q_{c,0} = \frac{1}{D_{c,0}} \sum_{d=1}^{D_{c,0}} \mathbb{1}(m_{dc,0} < 9).$$

Directional expectation: A higher share of districts with less than nine seats is expected to be associated with a larger average gain for smaller parties.

The nine-seat cutoff reflects the 10% natural quorum benchmark discussed in the theoretical background. Because of the limited number of cantons in the main-sample and their differing rules, this relationship is interpreted descriptively.

3.2.2 Exploratory question 2: Explicit electoral thresholds

The implementation of explicit electoral thresholds could limit the impact of the Double Pukelsheim Effect on smaller parties. Explorartory question 2 therefore analyses whether the mechanical seat-share gains differ between elections with and without an explicit quorum. An explicit electoral threshold, also referred to as an explicit quorum, is a legally specified minimum vote share that a party or list group must reach at a certain level to qualify for seat allocation. The different implementations are stated in Section 2.1.4. The comparison uses the difference between the two groups' unweighted mean seat-share gains.

$$D_Q = \bar{\Delta}_{\text{without quorum}} - \bar{\Delta}_{\text{with quorum}}.$$

Directional expectation: On average, the gain of the aggregate smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation is smaller in elections with an explicit electoral thresholds than in elections without one.

This comparison is interpreted descriptively because the level and geographical application of the thresholds differ across cantons and elections and coincide with other institutional differences.

3.2.3 Exploratory question 3: Change across successive elections

The main hypothesis examines the average mechanical seat-share difference across elections. The following analysis extends that comparison by examining the persistence and the magnitude of the effect within cantons. For this the elections are numbered chronologically from each canton's first election under DP. Here $\Delta_{c,t}$ describes the seat-share difference from the main hypothesis. The c is the canton identifier and t stands for the election number. Thus the change between consecutive elections is:

$$g_{c,t} = \Delta_{c,t} - \Delta_{c,t-1}, \qquad t \geq 2.$$

Stable positive values of Δ_e indicate a persistent mechanical advantage for smaller parties and positive values of $g_{c,t}$ indicate an increased advantage compared to the prior election. The within-canton perspective provides a separate image for each canton and prevents the mixing of different electoral systems. For the analysis, no directional expectation is derived since the smaller parties' seat share, the party composition, as well as the district magnitudes and electoral rules can change between elections.

4 Research Design and Analysis

This section outlines the research design and methodological approach to answer the research question. First the counterfactual strategy is explained, before the counterfactual assumptions, data processing, measures and finally the results are presented.

4.1 Counterfactual Strategy

Like Leuzinger (2018a; 2018b) and Pukelsheim & Schuhmacher (2011) uses this thesis a counterfactual reallocation design. The main analysis covers all 31 cantonal parliamentary elections held in the eight main-sample cantons from Hagenbach-Bischoff to Double Pukelsheim. The two additional elections from Graubünden are included as contextual comparisons since the canton previously used a majority voting system. At the computational level, the observations are electoral lists within electoral constituencies and elections, while the primary unit of comparison is the election.

Each election is evaluated under two allocation regimes. The observed outcome is the official allocation under Double Pukelsheim, obtained from the relevant cantonal election authority. The counterfactual outcome is obtained by applying the specified former legal allocation rules to the same election's observed votes, electoral units, seat magnitudes, and reported list structure. Each election therefore serves as its own comparison.

The process to obtain the counterfactual was split into five steps.

First, the official election files are imported and standardized to obtain, for every election, electoral unit, and reported list or list group, the observed votes, seat magnitude, and official Double Pukelsheim seats. Party and list labels are harmonized and voter numbers or vote shares are if necessary reconstructed.

Second, the counterfactual inputs are defined by holding the observed votes, electoral units, seat magnitudes, and reported list configuration constant. Joint lists reported as one unit in the official results remain pooled while lists affiliated with the same party but running independently remain separated.

Third, the specified former allocation regime is applied separately within each electoral unit. All counterfactual allocations use the district-level Hagenbach–Bischoff method described in Section 2.1.2. Where the former cantonal rules contained additional rules such as thresholds, majority-clauses, or tie-breaking provisions, these are incorporated into the calculation. The applicable canton-specific rules and features are documented in Section 4.2.

Fourth, the counterfactual seats obtained in the individual constituencies are aggregated to the party-election level and merged with the official Double Pukelsheim seat totals.

Finally, the resulting datasets are validated by checking party coverage, vote and seat consistency, electoral-unit seat totals, and the total number of seats allocated under both regimes. Where the source data provide multiple measures of the official Double Pukelsheim result, such as upper-apportionment or elected-candidate counts, these measures are additionally compared. If exact ties between parties occur, they are if possible resolved with the applicable former cantonal laws. For the two cases in the analysis in which the tie persisted, both possible seat allocations were presented for the affected parties. This assures that a seat is never assigned unintentionally. The two mentioned unresolved alternatives were observed in Graubünden 2022 and Schwyz 2024 and concerned only parties above the 10% threshold; they did therefore not affect the main smaller-party comparison.

With this methodology, the resulting contrast between the observed Double Pukelsheim allocation and the counterfactual is interpreted as a mechanical allocation difference conditional on the observed post-reform votes and not as the reform's complete causal effect.

4.2 Counterfactual Assumptions and Scope

Applying the general counterfactual strategy of this thesis requires several canton-specific decisions concerning electoral rules, list alliances, electoral units, and seat magnitudes. This subsection explains these decisions and defines the scope of the comparison.

Across the nine cantons examined, the counterfactual applies a district-level allocation to the electoral units retained from each observed election. In eight cantons, this generally reconstructs the formerly applicable Hagenbach–Bischoff regime, subject to the exceptions summarized in Table 2.[3] In Graubünden, which previously used majority voting, the Hagenbach–Bischoff allocation is entirely hypothetical and is therefore excluded from the main transition estimate. In every case, the observed votes, electoral-unit seat magnitudes, and reported list structure are held constant. The common Hagenbach–Bischoff procedure is described in Section 2.1.2. This subsection therefore focuses on canton-specific adaptations and limitations of the reconstruction.

[3] Aargau (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b); Staatskanzlei des Kantons Aargau, 2004, section "Neu: 140 statt 200 Mitglieder"); Graubünden (Kanton Graubünden, 2003/2018, Art. 27; hypothetical HB specification based on Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)); Nidwalden (BGE 136 I 352, E. 3.2); Schaffhausen (Regierungsrat des Kantons Schaffhausen, 1979, §§ 2, 36–37; Regierungsrat des Kantons Schaffhausen, 2007, pp. 2, 6); Schwyz (Urteil des Bundesgerichts 1C_407/2011 vom 19. März 2012, E. 5.2); Uri (Kanton Uri, 2018, Art. 22–23, "Bisher"; Kanton Uri, 2019, pp. 4, 11); Valais (BGE 140 I 107, Sachverhalt, pp. 108–109; Kanton Wallis, 2004a, Arts. 154–156); Zug (Kanton Zug, 2006, §§ 46–47, 55–56; BGE 136 I 376 E. 4.2, 4.4 and 5.1); and Zurich (Bentz, 2003, p. 2; Kanton Zürich, 1983/2005, §§ 87–88b)

Importantly, the comparison in this thesis concerns the complete election-specific allocation regimes previously used in the cantons, rather than only the mathematical difference between the two apportionment algorithms. The observed outcome incorporates the Double-Pukelsheim rules applicable in the respective election, including any explicit electoral quorum. Further the counterfactual applies the relevant former district-level rules to the observed votes and electoral structure. Consequently, differences in seat allocation may reflect both the change in apportionment method and associated differences between the applicable rule packages.

Table 2: Former electoral systems and counterfactual treatment by canton.

Canton	**Pre-reform allocation**	**Counterfactual treatment**	**Canton-specific counterfactual rule**
AG	HB in 11 electoral districts	Former HB allocation reconstructed	Standard HB procedure
GR	majority voting	Hypothetical district-level HB allocation in 39 electoral districts ; excluded from the main estimate	Hypothetical standard HB specification based on the basic formula in Aargau Law
NW	HB in 11 municipalities	Former HB allocation reconstructed	Standard HB procedure
SH	HB in 5 multi-seat districts, majority rule in 1 single-seat district	Former mixed district-level system reconstructed	Ceiling-based distribution number; relative majority in the single-seat district; statutory tie-breaking where required
SZ	HB in 30 municipalities	Former HB allocation reconstructed	Standard HB procedure; unresolved exact ties retained as alternatives
UR	HB in 8 municipalities, majority voting elsewhere	HB reconstructed for the 4 municipalities using DP	Tie-breaking by initial remainder and then party votes; unresolved alternatives retained
VS	HB in 14 districts	Former district-level HB regime reconstructed	8% district eligibility threshold: ineligible lists' votes remain in the distribution-number calculation
ZG	HB in 10 municipalities; majority voting in Neuheim	HB applied in all 11 municipalities; hypothetical for Neuheim	Standard HB procedure; hypothetical HB application in Neuheim
ZH	HB in 18 electoral districts	Former HB allocation reconstructed	Tie-breaking by party votes, followed by lot; unresolved lots retained as alternatives

Of all canton-specific provisions in Table 2 several do not affect the smaller-party outcomes. For example, the statutory tie-breaking rules in Schaffhausen, Uri, and Zurich were not triggered in any analyzed election. The Schaffhausen rule for calculating the distribution number, though, produced

differences in two cases: 5,022 rather than 5,023 in Klettgau in 2008 and 2,196 rather than 2,197 in Neuhausen in 2024. These differences changed neither the initial nor the final seat allocation. In Schwyz 2024 and in Graubünden 2022, an unresolved exact tie produced alternative HB allocations for parties above the 10% threshold. Thus it only affected individual party totals but not the aggregated seat total of smaller parties used in the main analysis. The relative-majority allocation in Schaffhausen's single-seat district was nevertheless applied and remains part of the reconstructed former system.[4]

Beyond the statutory allocation rules, the counterfactual requires an assumption about unobserved list alliances. Unlike Leuzinger and Pukelsheim & Schuhmacher, this analysis does not carry historical or hypothetical alliances forward. List alliances are in this thesis seen as strategic and election-specific. This means that election-specific, historical alliances cannot simply be assumed to persist across subsequent elections with different parties and vote distributions. Furthermore, the political landscape changed. In Zurich, for example, the last Kantonsrat election under Hagenbach–Bischoff took place in 2003. The most recent election in Zurich was held 20 years later in 2023; during that time new parties like the GLP emerged, others like the BDP disappeared and others might have undergone a political shift. (Statistisches Amt des Kantons Zürich, 2003, table 3; Amt für Statistik und Daten Kanton Zürich, 2026; Grünliberale Partei Kanton Zürich, n.d.). Therefore the counterfactual preserves joint lists or list groups represented as a single unit in the observed election data. But it does not reconstruct historical alliances or assume new alliances between separately reported units. This provides a transparent counterfactual that holds the observed electoral configuration constant and estimates the mechanical allocation difference between the specified election-specific regimes.

The exclusion of unobserved list alliances is an important limitation of the counterfactual. Under Hagenbach–Bischoff, list alliances can pool votes during seat allocation and may thereby improve the chances of representation for smaller lists (Pukelsheim & Schuhmacher, 2011, p. 1590). However, Vatter (2015) notes that in list alliances usually the largest partner profits, whereas sub-list alliances tend to benefit smaller parties. Sub-list alliances form a subgroup within a larger list alliance, whose members are initially treated together when distributing seats within that alliance. From the eight main-sample cantons, four allowed list alliances and two of these four explicitly allowed sub-list alliances.[5] Since the alliances which would have been formed are unkown, excluding them may overstate or understate the estimated gains under Double Pukelsheim for the four affected cantons.

A second research-design choice concerns the electoral units and their seat magnitudes. In most cantons, the reform retained the former territorial districts or municipalities, meaning that the observed electoral units correspond to those used under HB. If regarding the district magnitudes any changes occurred between two periods, then they were applied. The counterfactual therefore holds each observed election's electoral units and seat magnitudes constant across the two regimes.

The remaining deviations from this general approach are summarized below.

[4]These findings are based on the author's calculations of the official election data described in Appendix 1. Sources for the applicable legal provisions are provided in the preceding footnote.

[5]The classification refers to the electoral rules applicable immediately before the introduction of Double Pukelsheim. (Leuzinger, 2018a, p. 41, Table 1 ; Kanton Uri, 2018, Art. 13(1), "Bisher" column, p. 5). Sub-list alliances were explicitly permitted in Aargau (Kanton Aargau, 1988/2004, § 8) and Schaffhausen (Regierungsrat des Kantons Schaffhausen, 2007, pp. 14, 18).

ℹ Treatment of the Special Cases

Graubünden. The HB allocation is hypothetical and applied using the common allocation procedure employed elsewhere in the analysis. Its basic formula is taken from the former Aargau law (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)). Thus the observations are shown in grey and excluded from the main transition estimate. In the 2022 election, an unresolved exact tie between the "FDP" and "SP_GRUENE" appears in Rheinwald; to resolve it a hypothetical lot would be needed. The tie, which does not affect the aggregated seat total of smaller parties, is therefore retained as an unresolved alternative in the counterfactual.(Kanton Aargau, 1988/2004, § 13; authors calculations using Kanton Graubünden, 2022) Robustness checks using Zurich's party-vote tie-breaker and Schaffhausen's ceiling-based quota preserved all possible allocations across the 39 districts in both 2022 and 2026 (Kanton Zürich, 1983/2005, § 88b; Regierungsrat des Kantons Schaffhausen, 1979, § 36).

Zug. Neuheim used majority voting before the reform. An exact majority counterfactual cannot be reconstructed because the results of a hypothetical second ballot are unobservable. Neuheim is therefore included using a hypothetical HB allocation.(BGE 136 I 376 E. 5.1; Kanton Zug, 2022a, 2022b, 2022c, 2026) The specific implementation approach adopted for Neuheim has no bearing on the main analysis.

Uri. Only Altdorf, Bürglen, Erstfeld, and Schattdorf used Double Pukelsheim in the elections analyzed here. They account for 37 of the Landrat's 64 seats (Standeskanzlei Uri, 2020, p. 1). Uri's vote shares, including the less-than-10% classification of smaller parties, are therefore based only on the four municipalities using Double Pukelsheim, not on the whole canton.

Valais. Upper apportionment is conducted separately within six electoral regions rather than canton-wide (Lüscher, 2024, p. 57).

Schaffhausen. The parliament was reduced from 80 to 60 seats when Double Pukelsheim was introduced. The counterfactual retains the 60-seat structure. (Regierungsrat des Kantons Schaffhausen, 2007, pp. 1-2, 6).

With the counterfactual rules and scope established, the next subsection describes the official election data and their standardization across cantons.

4.3 Data Sources and Processing

This subsection describes the official election data used in the analysis and explains how the heterogeneous cantonal source files were transformed into a harmonized dataset.

The official election results were obtained primarily from publicly available datasets published by cantonal state chancelleries and statistical offices. Where the data wasn't available online in structured formats, such as CSV or XLSX, they were requested from the responsible cantonal authorities directly. In the case of Schaffhausen, no such format was available; therefore the results were extracted from publicly available PDF files. The precise sources and elections covered are reported in Appendix Table 4. The granularity and organization of the source data varied considerably across cantons and elections. In Schwyz, for example, the 2016 results were distributed across 30 municipality-specific csv

files, whereas the 2020 and 2024 results were each provided in a single file. In total 110 workbooks or files were used to extract the data.

Party and list names were standardized using readable, clear and consistent abbreviations. The same political organization was assigned a common identifier across cantons and elections, such as “SVP”, “GLP”, “ALG”, and “PIRATEN”. The “CVP” and “BDP” remain separately identified before their merger, while “DIEMITTE” is used after the formation of the new party in 2021 (H.-P. Schaub, 2022). In bilingual Valais, French- and German-language labels referring to the same party were harmonized, generally using the German abbreviation for consistency with the other cantons. However, “SP_GAUCHE_CITOYENNE” and “SP_GRUENE_UNABHAENGIGE” were retained as distinct identifiers because they represented separately reported lists with partly different organizations in the background.

The treatment of joint and affiliated lists follows the electoral units reported in the official results. Where several political organizations contested an election on a single joint list, that list was treated as one analytical unit. Examples include “SP_GRUENE” in GR_22 and “FGA_AL_PDA” in ZH_23. It is important to state that joint lists and the list alliances used under Hagenbach-Bischoff differ fundamentally. When forming a joint list, parties run together as a single electoral list (party), while under list alliances the parties run separately and the votes of the alliance parties are pooled later for the seat-allocation. (Pukelsheim & Schuhmacher, 2011, p. 1590) In addition to joint lists, in some cantons also affiliated or special-interest lists ran as an independent list. This phenomenon occurred especially strongly in Schaffhausen, where, for example, in 2024, seven affiliated lists ran. (Kanton Schaffhausen, n.d.). Since these lists entered the official Double Pukelsheim allocation separately, they were also retained as separate units in the analysis. This decision possibly affects the aggregate seat share of smaller parties.

Beyond the harmonization of party and list names, the cleaning process involved restructuring vote data, identifying the relevant electoral units, and assigning their election-specific seat totals. The source files were converted into comparable tables containing party votes and the seat magnitude of each electoral district. Where results like list totals were repeated, the scripts first checked that these repeated values were consistent and then retained one observation per municipality and list, preventing the totals from being counted multiple times. The geographic preparation depended on the structure of the official files and was widely different across the cantons. In Zurich for e.g municipal party votes were summed within the electoral districts identified by the source variable *Geschaeft* while Aargau’s files already contained district totals which were selected using the *DISTRICT* classification. While in Valais, the 14 sub-districts were assigned to six electoral regions in R according to the official electoral structure, preserving the different geographic levels required for the sub-district-level HB counterfactual and the electoral district-level DP upper apportionment.

District seat magnitudes were obtained from the available official information for each election. They were either already stated in the files, reconstructed with the files’ data or if necessary entered manually from official cantonal seat-allocation records .

The shared finalization function then produced one observation per standardized party or list code and election. It created an election identifier combining the canton abbreviation and election year, such as *ZH_23*, and joined the HB and DP seat totals by party. The final columns were *election_id*,

parties, *vote_share*, *small_party*, *seats_hb*, *seats_dp*, *hb_tie_note*, *parliament_size* and *seats_allocated_dp*. The last two variables distinguish the full parliament from the seats covered by the comparison, this because of the Canton Uri, where these values are 64 and 37, respectively. The column *hb_tie_note* contains explanations of unresolved ties. The priorly used district identifiers, raw party votes, and district magnitudes served as intermediate inputs and were not included in the final combined dataset.

Vote shares were derived using the information available in each canton's official election files. For Aargau, Zurich, Zug, Valais, Uri and Schwyz 2016 and 2020 elections, party votes were divided by the corresponding district's seat magnitude to obtain voter equivalents. Based on the applicable law, were the quotients either rounded to the nearest integer or kept unrounded. The equivalents were summed by party across constituencies, and each party's vote share was then calculated as its aggregated voter number divided by the total voter number of all included parties, multiplied by 100. For the remaining cantons, the voter equivalents or a share based on it was given in the respective files. The shared finalization function then retained these vote shares.

The Double Pukelsheim results were assembled separately within the canton-specific scripts from official seat tables, municipal allocations, or elected-candidate records. Each script supplied its election-specific DP results to the shared finalization function, which joined them to the HB counterfactuals by standardized party identifier. Validation routines checked electoral-unit coverage and seat totals and, where alternative representations were available, reconciled candidate counts or aggregated lower allocations with reported list mandates or upper-apportionment totals. District-level consistency checks and final allocation totals were applied according to the available source structure. Uri's allocation checks covered the 37 proportional seats included in the analysis. Unresolved Hagenbach-Bischoff ties were retained as alternative allocations.

One of the consistency checks revealed an error in the Zurich 2011 election data. There the party-votes total for the "SP" was doubled across all constituencies. The mistake was found through the comparison of the summed party votes with the reported unit totals and cross-checking candidate and supplementary votes. Followingly, the error was reported to the Statistical Office of the Canton of Zurich. They confirmed the mistake and resolved the underlying cause, which was that a duplicate "SP" entry in the list-to-party mapping table had caused double counting during the aggregation of the official municipal results. The corrected dataset was uploaded and is used in this analysis. (S.Graf, personal communication, 2026-09-11)

All processing was implemented in the canton-specific R scripts in *data_cleaning_cantons*, supported by the shared functions in *00_hb_allocator.R*. Each *process_canton_*()* function combined its cleaned elections into a cantonal dataset. These nine datasets were subsequently appended row-wise using *bind_rows()* to create *all_elections*, preserving one observation per standardized party or list and election. The combined dataset contains all 33 elections, while the Graubünden elections are excluded from the main-sample summaries. Detailed election data sources and legal bases are documented in Appendix 1, with the cleaning code provided on the GitHub repository and on the HSG Thesis Management Platform.

4.4 Measures and Analytical Strategy

The *analyzed_area* comprises the whole canton except in Uri, where it covers only Altdorf, Bürglen, Erstfeld, and Schattdorf, the four municipalities using Double Pukelsheim. The variable

seats_allocated_dp records the number of seats covered by the comparison, while *parliament_size* records the full size of the cantonal parliament. These values are equal in all analyzed cantons except Uri.

For the purposes of this thesis, a smaller party is defined as an electoral list or list group with a vote share below 10% in the analyzed area. Youth lists and other affiliated lists are treated as separate analytical units whenever they enter the official Double Pukelsheim allocation separately. Accordingly, the term *party* refers to the standardized electoral list or list group recorded in the dataset and does not necessarily correspond to a broader organizational party family. *Smaller parties* are identified separately for each election using their observed vote shares. The same set of smaller parties is then used when comparing the two seat allocations within that election.

The main analysis compares the combined seat shares of smaller parties within each election. For this purpose, the seats of all lists below the 10% vote-share threshold are summed separately under Double Pukelsheim and Hagenbach–Bischoff. Each total is divided by the number of analyzed seats, recorded in *seats_allocated_dp*, and multiplied by 100 to obtain the corresponding seat share as a percentage. This produces one aggregate seat-share difference per election. Using *seats_allocated_dp* ensures that the results for Uri are not distorted.

The election-level outcome is the difference between these two seat shares, denoted by Δ_e and defined in the main hypothesis. Positive values indicate that smaller parties collectively receive a larger seat share under Double Pukelsheim while negative values indicate a smaller share. A negative or positive aggregate difference does not mean that each included party loses or gains seats respectively.

The main hypothesis is assessed using the average seat-share difference aross the 31 elections in the main sample. In that analysis every election receives equal weight, which means that cantons with more elections held under DP are influencing the average more strongly. Since the parliament sizes differ, a supplementary weighted mean is added, where more weight to bigger parliaments is given. To better display the variation and typical gain, the median, the first and third quartiles are reported additionally. The main summaries show Graubünden only contextually for the reasons set out in Section 4.2.

The 10% threshold analysis is complemented by sensitivity analyses with 8% and 12% as a cutoff-value. Moreover is a supplementary analysis examining the mechanical seat-share gain of individual parties and grouping them into groups within the same vote-share range to analyse parties with which electoral support profit how much.This analysis provides context for the aggregate findings.

All three exploratory questions are descriptive since they examine differences across institutional settings and successive elections without isolating the causal effects of individual electoral rules.

For EQ1, election-level seat-share gains are first averaged within each canton. These canton-level outcomes are examined against mean district magnitude and the share of districts below nine seats. These two measures are calculated from the same fixed allocation in each cantons last pre-reform election and with the Schaffhausen parliament adjusted to 60 seats. These historical characteristics are held fixed when comparing the canton-level average gains across the analyzed elections. Uri covers the same four municipalities included in the HB–DP comparison, and Valais's district count refers to its 14 Hagenbach-Bischoff allocation districts. Both district measures are calculated once per canton from this baseline and are held fixed across its analyzed elections. For each measure, cantons are grouped

below or at or above its unweighted mean across the eight main sample cantons. Group means give each canton equal weight, and the group difference is calculated as below-mean minus at-or-above-mean. Graubünden is shown only for context. The historical reference year applies to all district measures. Historical seat allocations are documented in the cantonal publications and court judgments cited below.[6] Later district magnitudes may change, so the district measures describe associations with the initial structure. For Schaffhausen, the 2004 structure is adjusted from 80 to 60 seats using the published hypothetical allocation in the government's reform report (Regierungsrat des Kantons Schaffhausen, 2007, Table 1, p. 2). The election-specific Hagenbach-Bischoff to Double Pukelsheim allocations are unchanged.

For EQ2, elections are grouped by whether or not an explicit electoral quorum applied under Double Pukelsheim. Within each group, the mean of Δ_e gives each election equal weight, so cantons with more observed elections contribute more. The group difference is calculated as the mean without a quorum minus the mean with one. Graubünden is shown for context and excluded from both means. Since the observations differ structurally in terms of quorum levels, geographical scope and others, the analysis is only descriptive.

To examine successive elections, the seat-share difference Δ_e is plotted separately for each canton and its successive elections. The aim is to analyze whether the mechanical seat-share gain persists across elections. To identify possible influencing factors, they are noted next to the election in which they occurred. Later changes and fluctuations are interpreted with the help of the additional context provided by the notes. Besides the separate analyis of each canton, the results are also set into the between-canton context to possibly identify general patterns.

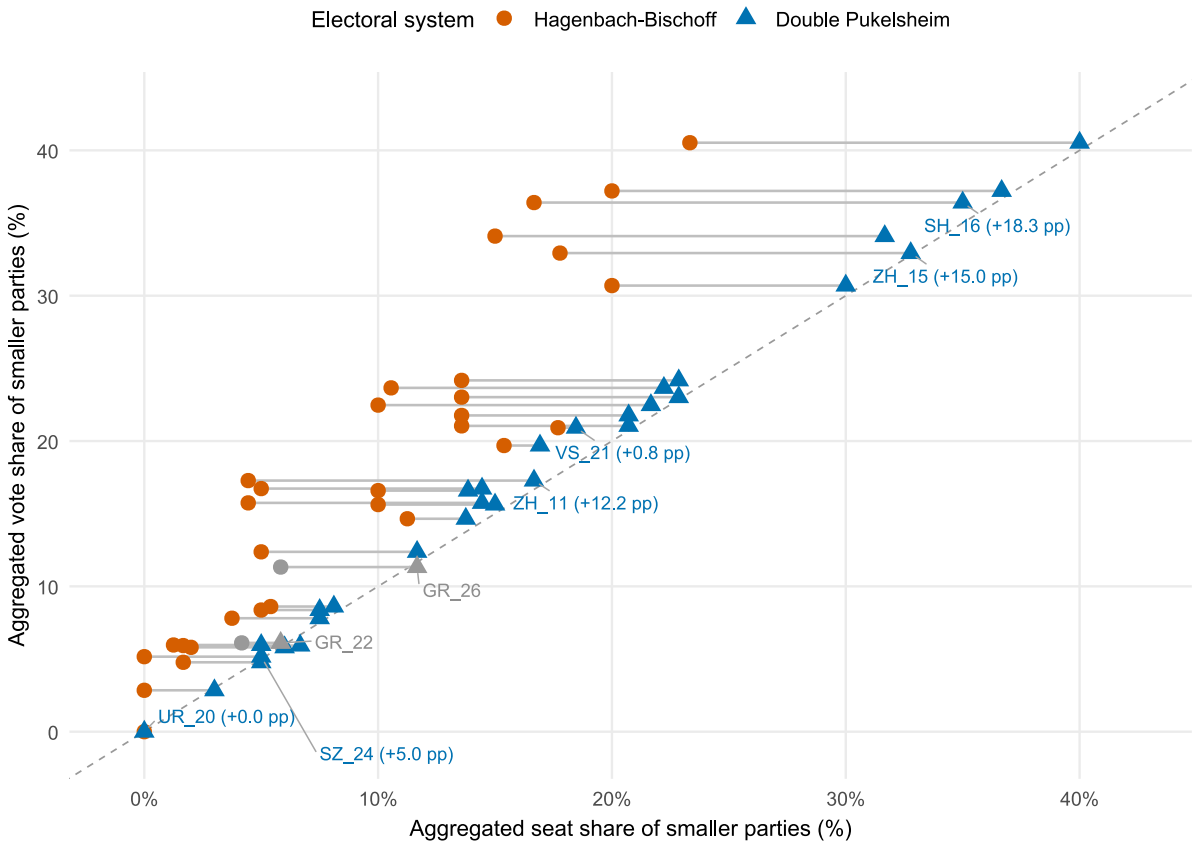
4.5 Results

4.5.1 Main Hypothesis: Smaller parties' seat-share difference

Figure 1 compares the aggregate smaller parties' combined seat shares under the observed Double Pukelsheim allocation and the counterfactual Hagenbach-Bischoff allocation for every available election. Smaller parties are defined as parties with less than 10% of the vote share in the analyzed area. This seat-share difference allows comparisons across elections with different numbers of analyzed seats.

[6] Sources by canton and reference year: Aargau 2005 (Kanton Aargau, 2004); Graubünden 2018 (Kanton Graubünden, n.d.); Nidwalden 2010 (BGE 136 I 352, E. 3.5); Schaffhausen 2004, adjusted to 60 seats (Regierungsrat des Kantons Schaffhausen, 2007, Table 1, p. 2); Schwyz 2012 (Urteil des Bundesgerichts 1C_407/2011 vom 19. März 2012, Sachverhalt A); Uri 2016 (Standeskanzlei Uri, 2016, p. 5); Valais 2013 (BGE 140 I 107, Sachverhalt A); Zug 2010 (Staatskanzlei des Kantons Zug, 2010, p. 92); Zurich 2003 (Regierungsrat des Kantons Zürich, 2002).

Figure 1: Combined seat shares of smaller parties under HB and DP



Grey lines connect the two allocations for the same election. The dashed diagonal marks equality between aggregate vote and seat shares. Points above it indicate underrepresentation; points below it indicate overrepresentation.

i Main Hypothesis summary

Across all 31 transition elections (Graubünden excluded), the unweighted mean gain in smaller parties' share of analyzed seats was 7.6 percentage points (Q1: 3.5; median: 6.7; Q3: 10.8). The mean weighted by each election's number of analyzed seats was 7.8 percentage points.

These results descriptively support the main hypothesis and indicate that smaller parties indeed in the 31 elections receive a larger share of the analyzed seats under Double Pukelsheim than under Hagenbach-Bischoff. This effect is partically evident through that in every election where small parties contested (every election but Uri 2020), they aggregately received a larger seat share. This finding is in accordance with Leuzinger's observation that smaller parties tend to profit from the transition (Leuzinger, 2018a, p.42). The weighted mean, unweighted mean and median are all relatively close to each other, indicating that the effect is not driven by a few elections with many seats and that the positive effect isn't determined by extreme outliers.

The average gain differs across the cantons. For example the Canton of Schaffhausen has a stable high seat gain for smaller parties. There the high seat-share gain may be partially explained through the decision to treat affiliated parties as independent lists. Therefore these affiliated lists were classified as smaller parties if they didn't reach the 10% threshold themselves and with that increase the relative gain for smaller parties. When the affiliated parties were attributed to their mother parties, the gain

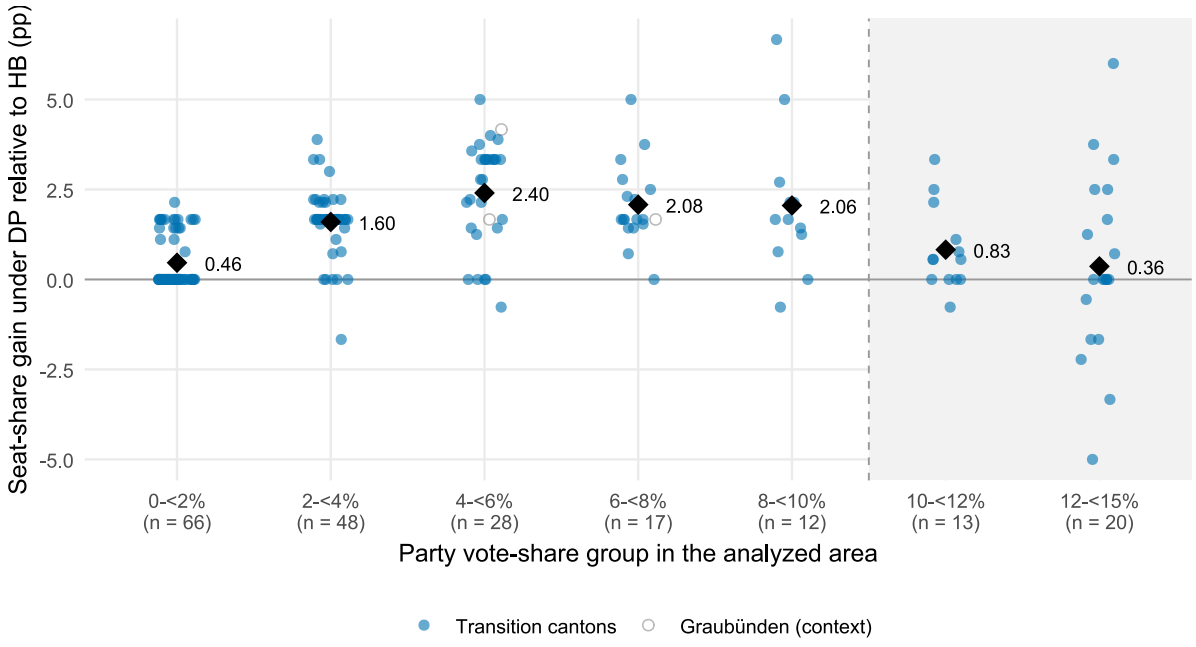
dropped from 15.67 to 8.67 percentage points. No matter the treatment of the affiliated lists, are the gains in Schaffhausen comparatively high. By contrast, the gains were constantly relatively low in Uri and Schwyz, where on average 0.5 and 1.66 smaller parties contested per election.

The main finding, that smaller parties aggregately benefit from the transition, is robust to alternative definitions of smaller parties. Using a vote-share cutoff of less than 8%, the mean gain is 6.77 percentage points. The smaller parties also collectively receive a higher seat share under Double Pukelsheim in 28 of the 29 elections where at least one party with a vote-share below 8% contested. The exception is VS 2021, where the aggregated difference is zero and the upper-allocation process was not done across the whole canton. With a cutoff of less than 12%, the mean gain is 7.91 percentage points. Regarding the collectively higher seat share in this case, in 30 of the 30 relevant elections, the smaller parties received a higher seat-share gain. In Uri 2020, no party received less than 12% of the vote, so this election is excluded from the count of elections with qualifying smaller parties.

As a further sensitivity and robustness check, the results were tested with a leave one canton out check. In that check the means are recalculated with each canton being omitted once. The positive seat-share gain persisted across each check. This persistance fits the finding that in 30 of 31 elections of the main sample, a positive aggregated seat-share gain for smaller parties was observed. The detailed results for the checks can be found in the Appendix in Section 8.3.

The prior measures deliver a profound picture over the aggregated seat-share gain of smaller parties, but they do not reveal whether smaller parties profited independently, and which category of smaller parties profits most from the transition. This is analyzed in the following subanalysis.

Figure 2: Mechanical party-level seat-share gains for different vote-share groups



Notes: Each point represents one party-election observation. Black diamonds and labels show group means for transition cantons (pp); n counts these observations. Grey shading marks vote-share groups outside the smaller-party definition (10% or above).

The analysis shows that the vast majority of the smaller parties below 10% not only profit aggregately from the transition but also independently. The summary of the 204 party-election observations shows

that parties between 4-6% vote share tend to profit the most from the implementation of the Double Pukelsheim. For parties with more than 10% the seat-share gain is on average markedly lower and reaches for parties above 12 and below 15% a point where the parties with 0.46 percentage points barely profit anymore from the implementation. Also parties on the other side of the scale with less than 2% only benefit slightly; there 45 of 66 observed parties receive no seats under either allocation. Overall, the analysis shows that smaller parties tend to profit from the implementation whereas bigger parties on average subsequently lose their prior advantage under the old system. To be able to identify a possible gradual pattern, the analysis was additionally done without grouping. This additional analysis shows no clear direct relationship; further is there substantial variation between parties of the same vote share. Another observation is that parties with a very low share cluster around zero, while positive gains occur across most of the remaining range. Furthermore, most negative observations appear in the upper end of the range. The figure can be found in the Appendix in Section 8.2.

To sum up the analysis of the main hypothesis: The main sample covers all 31 cantonal parliamentary elections held under Double Pukelsheim in Switzerland from the cantons that transitioned from Hagenbach-Bischoff. Across these elections, smaller parties received under all definitions of smaller parties a higher aggregate seat share under Double Pukelsheim. This pattern also persisted throughout almost every within election comparison of smaller parties. Furthermore, the analysis showed that most smaller parties also profit on an individual basis from the transition. These findings are based on the mechanical allocation difference conditional on the observed votes and specified counterfactual rules and therefore do not capture the whole transition effect. The next section examines whether the magnitude of this effect is associated with 'the cantons' district magnitudes.

4.5.2 Exploratory question 1: Differences by district magnitude

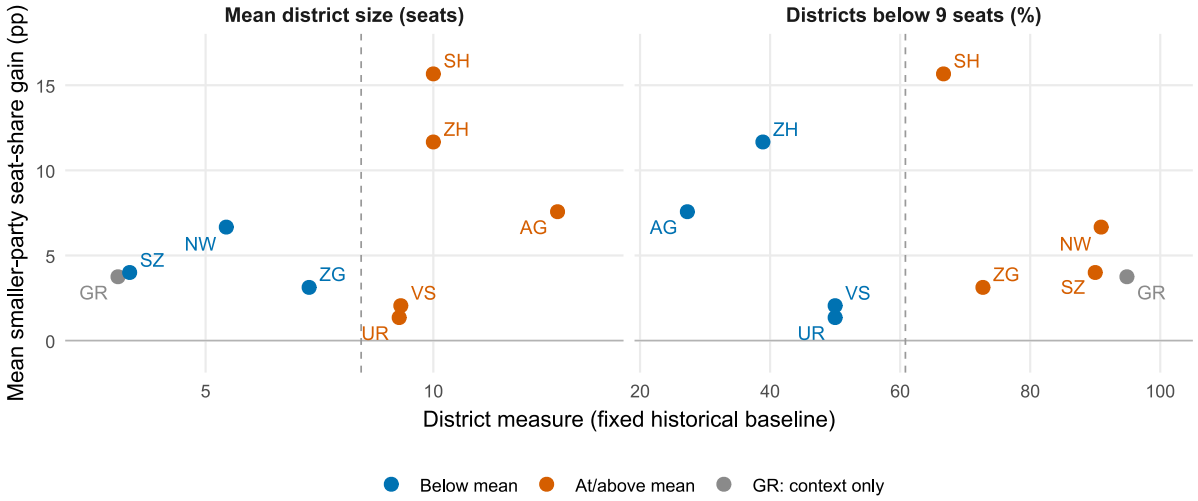
This section examines historical district structure using mean district magnitude and the share of districts with fewer than nine seats. Table 3 reports these measures alongside district counts and reference years. Both measures refer to the fixed historical baseline defined in Section 4.4, with Schaffhausen adjusted to 60 seats. Graubünden is shown for context only.

Table 3: Pre-reform district structure by canton

Canton	Districts	Reference year	Mean seats	Below 9 seats (%)
AG	11	2005	12.73	27.3
GR	39	2018	3.08	94.9
NW	11	2010	5.45	90.9
SH	6	2004	10.00	66.7
SZ	30	2012	3.33	90.0
UR	4	2016	9.25	50.0
VS	14	2013	9.29	50.0
ZG	11	2010	7.27	72.7
ZH	18	2003	10.00	38.9

Figure 3 relates each district measure to the canton-level mean gain in smaller parties' seat share. The cutoffs are defined by the mean of the eight relevant cantons and are the following: 8.42 seats for the unweighted mean magnitude and 60.81% for the share below nine seats.

Figure 3: Smaller-party seat-share gains by pre-reform district structure



Notes: Dashed vertical lines show the unweighted means across the eight transition cantons. Uri covers four DP municipalities.

EQ1 summary

For the variable mean district magnitude, smaller parties' mean seat-share gains were 4.60 percentage points in the below-mean group and 7.66 percentage points in the at-or-above-mean group, giving an absolute difference of 3.06 percentage points. For the share of districts below nine seats, the corresponding gains were 5.66 and 7.36 percentage points, an absolute difference of 1.70 percentage points.

Figure 3 shows no clear, consistent relationship between historical district structure and smaller parties' mean seat-share gains across the eight main sample cantons. For mean district magnitude, the group difference is opposite to the expected direction. For the share of districts below nine seats, it is consistent with the expectation but sensitive to Schaffhausen's inclusion. The numerical comparisons below give each canton equal weight.

To analyze the sensitivity of the results to certain cantons, a leave one canton out check was carried out. The check showed that the results in exploratory question 1 are sensitive to the omission of the canton of Schaffhausen. The omission of Schaffhausen lets the mean difference change from −3.06 to −1.06. For the share below 9 seats measure the direction switches with the omission of the canton. This means that without the canton of Schaffhausen, the average gains for the below mean groups are higher than for the above mean group, which oposses the estimated direction. Generally there is substantial variation in both groups, which means that also the omission of cantons like Uri and Valais has substantial impact on some of the measures. The exact results to the robustness checks can be found in Section 8.3.

Taken together, the two district measures reveal no consistent pattern linking the initial district structure to smaller parties' gains. This is combined with substantial variation within the analyzed groups. Further are the two measures sensitive to the omission of single cantons. The sensitivity supports the assessment of a missing pattern. The sensitivity to Schaffhausen can be linked to its comparatively high seat-share gain and possibly to the priorly discussed separate classification of affiliated lists as smaller parties. These descriptive comparisons therefore can not show any causal effect of the district magnitude.

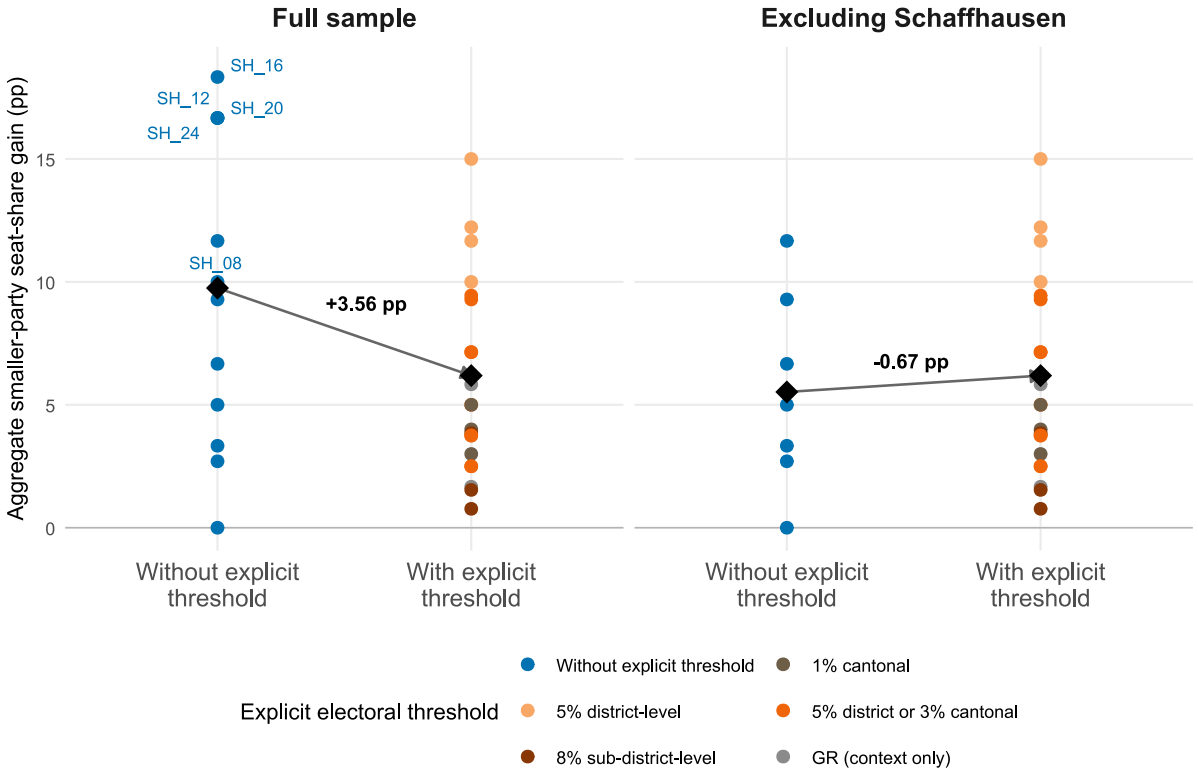
4.5.3 Exploratory question 2: Explicit electoral thresholds

Exploratory question 2 researches whether the implementation of explicit electoral quroums has an influence on the smaller parties seat-share gain. The explicit thresholds ensure that only parties which reach a specific vote-share are considered for the allocation of seats. This may prevent smaller parties' from entering the parliament, which is why it is expected that in cantons with a quorum, the seat-share gain of smaller parties is reduced.

As documented in the Table 1, do five out of eight main-sample cantons apply an explicit threshold. Of these five cantons, Aargau is a special case, since the canton applied it only after their first election under Double Pukelsheim. The most commonly used variant is the combination of a 5% district-level and a 3% cantonal-level threshold of which at least one has to be reached. This combination was used in 9 of 19 elections held with an explicit threshold.

The difference is visualized followingly as two grouped dot plots.

Figure 4: Smaller-party seat-share gains by explicit electoral threshold



Notes: Black diamonds show equally weighted election means, excluding Graubünden. Labels show the mean without threshold minus the mean with threshold (pp).

> **i** EQ2 summary
>
> Smaller parties' mean seat-share gain was 9.75 percentage points across 12 elections without an explicit threshold and 6.19 percentage points across 19 elections with one. The without-minus-with difference was +3.56 percentage points. Excluding Schaffhausen reverses the difference to −0.67 percentage points. The full-sample comparison is consistent with the expected direction, but is sensitive to Schaffhausen's inclusion.

Especially interesting is the case of the canton Aargau, where in 2009 the election was held without a threshold and in the elections 2012, 2016, 2020 and 2024 with a threshold. The data shows that the seat-share gain for the pre-reform election 2009 and the first post-reform election 2012 is equally big with 9.3 percentage points. In the subsequent elections, though, the seat-share gain dropped accordingly to the assumption. While the unchanged gain between 2009 and 2012 contrasts with the assumption, the subsequent elections are in line with it. For Aargau there is therefore no clear pattern visible; moreover, might the seat-share gain difference be influenced by further unobserved circumstances.

To analyze whether the effect holds despite the variation, the leave one canton out sensitivity check was performed. As seen in the summary above, is the analysis sensitive to the omission of Schaffhausen. Schaffhausen accounts for five of the twelve elections in the group without an explicit quorum, with all five having seat-share gains above the group average. The exact results for the omission of every canton are outlined in Section 8.3.

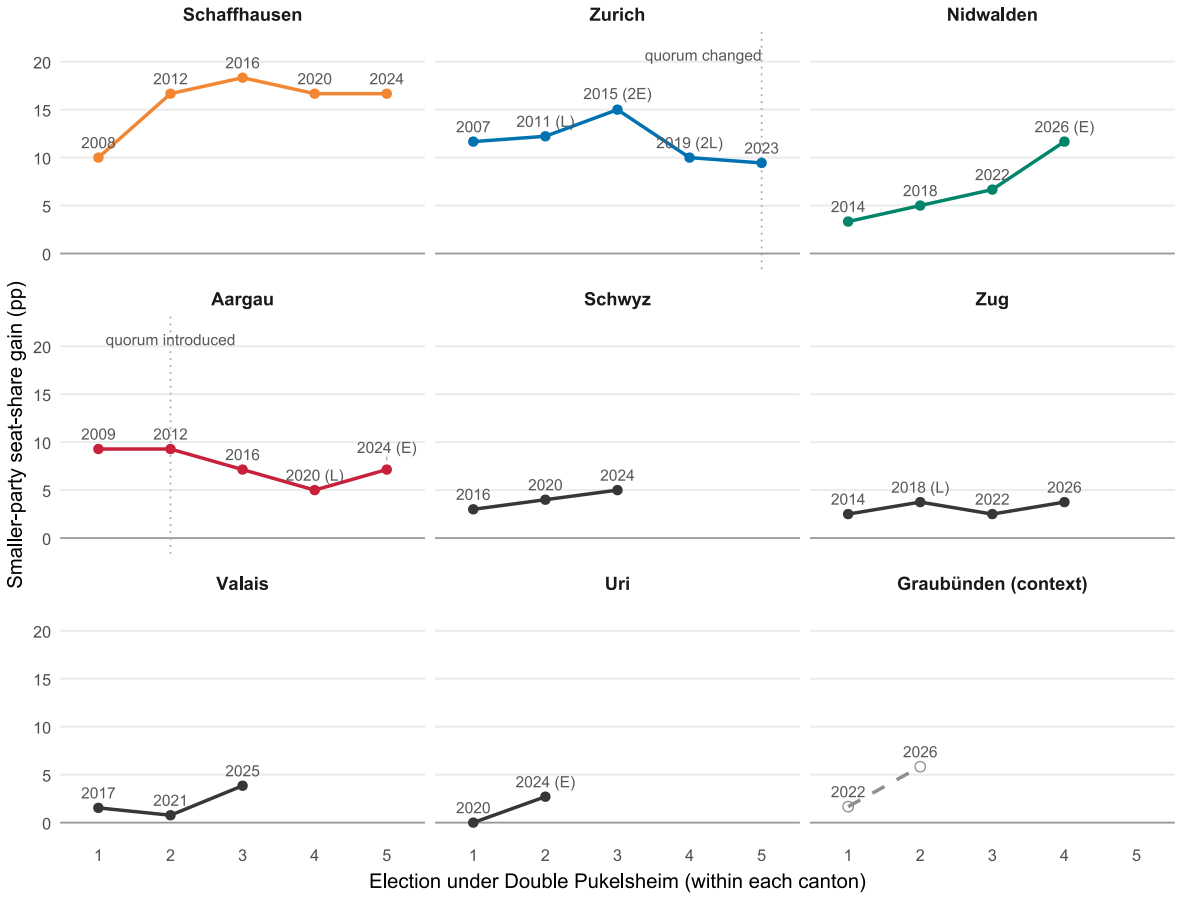
To summarize the analysis about the effect of the implementation of explicit quoroms, it can be stated that within the 31 relevant election a difference which matches the expectation can be seen. This difference though is sensitive to the inclusion of Schaffhausen. Furthermore, do the limited amount of elections held till date and the different application of thresholds and electoral law encourage to interpret the difference with caution.

4.5.4 Exploratory question 3: Change across successive elections

The preceding analysis showed that cantons differ in their implementation of the Double Pukelsheim system. These institutional differences motivate a closer examination of within-canton patterns. The last exploratory question therefore examines how the smaller parties' seat-share gains changed across successive elections within a canton. For that each canton and its successive changes are analyzed separately. To possibly better understand changes, events such as electoral reforms and changes in the party composition are documented. Generally stable positive values of Δ_e indicate a persistent mechanical advantage for smaller parties, while substantial fluctuation or even changes in sign would suggest that the effect is strongly dependent on the individual circumstances of an election.

Since Figure 5 aligns every Double Pukelsheim election for each canton independently by their order, the first election under the system is numbered 1, the next is numbered 2, and so on. This means that the same election number represents different calendar years across cantons. The same measures as in the main hypothesis are taken to calculate the seat-share differences. The smaller-party category is recalculated for each election using the existing vote-share cutoff of less than 10%.

Figure 5: Smaller-party seat-share differences across successive DP elections



Notes: Differences equal observed DP minus counterfactual HB combined smaller parties seat shares. Dotted vertical lines mark electoral rule changes. E/L: one parliamentary party enters/leaves the below-10% group; 2E/2L: two parties. Graubünden is shown for context only.

i EQ3 summary

The average within-canton change in smaller parties' seat-share gain was +1.6 pp from election 1 to 2 (8 cantons); +1.0 pp from election 2 to 3 (7 cantons); −0.5 pp from election 3 to 4 (5 cantons); +0.5 pp from election 4 to 5 (3 cantons). The average seat-share gains across the cantons were 5.2 pp in election 1 (8 cantons); 6.8 pp in election 2 (8 cantons); 8.4 pp in election 3 (7 cantons); 9.4 pp in election 4 (5 cantons); 11.1 pp in election 5 (3 cantons). For each transition, the average gives equal weight to all cantons observed in both elections. Graubünden is excluded.

Generally speaking, are the seat-share gains with some canton-specific fluctuations relatively persistent. As seen in the analysis of the main hypothesis, is there with *UR_20* only one election, in which no positive seat-share gain was seen. In the *UR_20* election, no party with less than 10% vote-share contested; thus no effect could be observed. Another interesting observation is that the average seat-share gain rises successively from election to election. The fact that the cantons of Schaffhausen and Zurich with the highest gains also experience drops while the average successive gain persists supports a possible effect regarding the rise of the gains. Additionally, is the analysis less dependent on political trends specific to a single election year, since for example the second elections held under Double

Pukelsheim range from 2011 in Zurich to 2024 in Uri. However, is the amount of cantons limited and further lowers throughout the later election. Moreover does the time range not isolate the pattern from political trends. A possible partial explanation for the observed pattern is the broader increase in party-system fragmentation, which was documented by Lüscher (2024, pp. 133-135). Since if smaller parties receive wider support, their combined disadvantage under the former Hagenbach-Bischoff and with that their seat-share gain under Double Pukelsheim may increase. Nevertheless, whether this mechanism actually explains the observed pattern remains untested.

Besides the broader image, there are also interesting patterns within the cantons. The biggest changes between two elections can be seen for Schaffhausen, Zurich and Nidwalden. In Schaffhausen the seat-share gain rose from the first to the 2nd election from roughly 10 to roughly 17 percentage points, where it settled afterward quite stably. A possible partial reason is the higher number of smaller parties in the parliament which increased from seven in 2008 to nine in 2012. From the 2015 Zurich to the 2019 Zurich election, the seat-share gain dropped by roughly 5 percentage points; a reason for this is possibly that both the GLP and the Greens exceeded the 10% smaller-parties threshold while the BDP lost its parliamentary representation. This led to the circumstance that from seven smaller parliamentary parties in 2015, the number dropped to four in 2019. In Nidwalden since the implementation the seat-gain share rose continuously. The strongest difference was observed from the 3rd to the 4th election, where the share rose by 5 percentage points. This effect might be partially influenced by the smaller party composition, since the green party fell below the 10% threshold. For Aargau, Zug, Schwyz and Valais we can observe relatively stable patterns with smaller fluctuations. The canton of Valais records relatively low seat-share gains across the three elections. This pattern is consistent with the possibility that the restricted pooling in Valais has a limiting effect. Nevertheless this analysis is not isolating this feature and a potential effect. For the canton of Uri only two elections were observed, which is why no pattern can be inferred.

In the two cantons Aargau and Zurich, the Double Pukelsheim system was adjusted after its implementation. In Aargau in 2012, a threshold was introduced (see Section 4.5.3). The assumption of the implementation of a threshold is that the smaller-party seat share gain declines with it. The post electoral change results for the canton of Aargau do not present a clear picture. There are results in accordance as also speaking against the assumption. The absence of a clear effect may be partly explained through that in each election the biggest list without parliamentary representation received only between 0.1% and 0.7% vote shares.[7] This suggests that the explicit thresholds in Aargau only had limited scope to exclude parties which otherwise would have won seats. In the canton of Zurich, in the year 2023 a quorum change applied. To the existing 5% district-level quorum an alternative 3% canton-wide threshold was added. This electoral change is in the tendency increasing the seat-share gain, since it possibly enables more parties to reach the seat-allocation process. In the 2023 election, though, no party profited from this clause, since every qualifying party also reached the 5% district-level threshold (Regierungsrat des Kantons Zürich, 2023, p.9). Further, the seat-share gain dropped from the 2019 to the 2023 election. Overall, the two electoral changes therefore had no visible direct effect on the smaller party seat-share gain.

[7]For cantonal vote shares and parliamentary representation, see T. Schaub (2012, p. 12), Staatskanzlei des Kantons Aargau (2016b, p. 2), Staatskanzlei des Kantons Aargau (2020, p. 2), and Staatskanzlei des Kantons Aargau (2024).

To check for robustness and sensitivity, the leave one canton out check was performed. The check was done for each election number separately. Throughout all checks does the seat-share gain remains positive. The mean change though from election three to four and from four to five switches signs when Zurich, respectively the canton of Aargau is excluded. These later transitions include only five respectively three cantons and are therefore more sensitive to individual cantons. The exact statistics for the robustness check can be found in Section 8.3.

The analysis across all 31 elections of the main-sample therefore indicates that the mechanical seat-share gain for smaller parties persists across successive elections. The strength and variation in the magnitude differ between the cantons. In the analyzed elections, the gain not only persisted, but rose across sucessive elections. The average within-canton change though isn't rising consecutively with a small decline between the third and fourth election. Moreover, the analysis is sensitive for the within-canton change to the omission of single cantons. The analysis and its effects may be partly affected through the changes in the party composition of the smaller-party group. The results therefore imply a consistent gain for smaller-parties across cantons and elections. However it can not be established and is not inferred or mentioned by the literature that additional elections under Double Pukelsheim increase the gain.

5 Discussion

In this paper, the research question: "20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?" was answered in two parts. In the first step, the mechanical effect on the smaller parties was examined. Subsequently the study examined which conditions could lead to differences in the strength of the mechanical effect across cantons or elections.

Regarding the first step, it was found that across elections the smaller parties collectively gain seat shares under Double Pukelsheim compared to Hagenbach Bischoff. The average aggregate unweighted seat-share gain was 7.6 percentage points. Further in 30 of the 31 main-sample elections, the smaller parties aggregately received a higher seat-share. The gain also persisted under the robustness tests. Further it was observed that most smaller parties also individually profited from the transition. The findings in the main hypothesis are consistent with the idea of the upper apportionment process that it lowers the share of ineffective votes and with that increases the equality of electoral success. (Lüscher, 2024, p. 3) The finding further matches the observations of the existing literature. Leuzinger states that in his analysis across seven elections, smaller parties generally received more seats under the new system (Leuzinger, 2018, p.42) Also in the Zurich 2011 election by Pukelsheim and Schuhmacher and in the Schwyz 2016 election analyzed by Lüscher, smaller parties profited from the implementation (Pukelsheim & Schuhmacher, 2011 ; Lüscher, 2024, p.13).

In the second step, it was examined whether the effect varied based on the different circumstances of the cantons. In EQ1 two measures analyzed whether the pre-reform district sizes have an impact on the seat-share gain or not. No consistent relationship between the prior constituency sizes and the gain for smaller parties was found. This is contrary to the formulated directional expectation. The expectation was further based on Lüscher's finding that in cantons which priorly had smaller districts, the proportionality had increased more strongly (Lüscher, 2024, pp.132-133). The lack of evidence in

this analysis does not contradict Lüscher's findings, since an improvement in overall proportionality does not need to have a direct impact on the seat-share gain for smaller parties.

A second exploratory question examined whether there is a difference in the mechanical seat-share gain between elections held with and without an explicit quoroum. The examination here observed a higher seat-share gain for cantons without an explicit quorum. This fits the literature, since Lüscher and the Federal State Chancellery both argue that explicit thresholds can restrict the gains among smaller parties. This since thresholds may partly offset the greater proportionality achieved through the upper apportionment (Bundeskanzlei, 2013, p. 18; Lüscher, 2024, pp. 107, 135). Despite the observed effect and the support of the literature for one, is the effect in this analysis sensitive to the inclusion of Schaffhausen. The sensitivity goes to a point that without the Schaffhausen elections, the effect reverses. Therefore the observed effect needs to be interpreted cautiously.

The third exploratory question analyzes the seat-share gains across successive elections within and with respect to each canton's electoral rules. Through this approach, it brings together elements of the preceding analyses. The results show that the positive seat-share advantage generally persists across successive elections, further supporting the main finding. Although average gains are higher at later election numbers, mean within-canton changes do not follow a consistently increasing pattern. Moreover, fewer cantons contribute to the later comparisons, and some changes are sensitive to the inclusion of individual cantons. The findings therefore do not establish that additional elections under Double Pukelsheim increase smaller parties' mechanical seat-share gain.

In order to analyze all relevant elections and their various underlying assumptions, a few assumptions which are detailed in Section 4.2 had to be made. The analysis holds all post-reform votes constant and identifies the conditional mechanical difference, without capturing the behavioural effect of the transition. The treatment of electoral lists further affects the interpretation since the exclusion of hypothetical alliances might, in the cantons which priorly knew such alliances, understate or overstate the smaller parties mandates under the HB counterfactual. However, is the magnitude and direction of this decision uncertain. Each affiliated list which ran independently was treated as a separate party, since the Double Pukelsheim algorithm also treats them separately. This distinction is relevant, since some of the affiliated lists would have otherwise not been in the smaller parties category. Finally, the limited number of cantons and their different electoral laws limit the extent to which the observed differences in gains can be attributed to a single feature.

For cantons considering a transition from Hagenbach–Bischoff to Double Pukelsheim, the following can be stated. With 30 of the 31 analyzed elections showing a collective seat-share gain for smaller parties, the findings suggest that a similar mechanical advantage would arise in other cantons adopting DP. However, its magnitude depends on canton-specific circumstances, such as electoral rules, district sizes, and vote distributions. Behavioural responses by parties and voters introduce further uncertainty. It therefore makes sense for cantons to calculate a hypothetical DP allocation based on their observed Hagenbach–Bischoff election results. This allows them to assess the potential mechanical seat-share changes while explicitly accounting for their electoral structure and the proposed DP rules.

Further research could extend this thesis in the following ways: With examining how plausible counterfactual list-alliances affect the estimated gains. Another starting point would be addressing the explicit quoroums and their possible effects. Specifically it could recalculate each observed election

under different electoral quorums, ceteris paribus, to isolate their mechanical effect on smaller parties' representation.

6 Conclusion

This paper examined the mechanical effect of the transition from the Hagenbach-Bischoff Method to the Double Pukelsheim on the seat distribution of smaller parties. The thesis focused on examining the seat-share differences between the two systems for parties with less than 10% vote share. For this all held 31 elections by the eight main sample cantons from HB to DP were analyzed, while holding votes constant.

For the elections of the main-sample, the average unweighted seat-share gain was 7.6 percentage points for the aggregate smaller parties group. In 30 of 31 elections, the smaller parties collectively gained seat-shares. The Uri 2020 election is the exception, where no list below the 10% smaller parties threshold contested. The result of a positive seat-share difference further persisted with alternative cutoffs for the smaller-parties group and when excluding individual cantons. These findings therefore descriptively support the main hypothesis.

The further analysis explored potential differences between cantons, based on pre-reform district sizes and explicit thresholds. For the relationship between the pre-reform district sizes and the seat-share gain, no consistent pattern could be observed. Regarding differences between elections held with and without a quorum, a higher seat-share gain for elections without a quorum was observed. However, the observed difference is sensitive to individual cantons. Further it was found that average gains persisted and even increased across successive election numbers, even when individual cantons were excluded. However the average within-canton changes were less consistent and sensitive to canton exclusions. This pattern does not establish that additional elections under Double Pukelsheim increase the seat-share gain.

However, this analysis is only focusing on mechanical differences conditional on the canton specific rules and on the observed post-reform votes and not estimating the reform's complete effect. This paper extends the existing knowledge about the mechanical seat-share gains for smaller parties by covering all 31 available elections for the eight main sample cantons, including the successive elections within each canton. This enables a wider basis to assess the magnitude and consistency of the smaller parties' advantage under the Double Pukelsheim.

7 Bibliography

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8 Appendix

8.1 Appendix 1: Explanation Data Cleaning Process

The table below identifies the election data and legal sources used in the canton-specific cleaning scripts and counterfactual Hagenbach–Bischoff allocation. It specifies source titles, election coverage, file types, and the purpose of publications used to reconstruct or validate the results, together with the relevant legal provisions. Clickable citations and source titles lead to online sources where available; unpublished material supplied by email is identified explicitly. Author–year and case citations correspond to entries in the bibliography.

Table 4: Election data sources and legal bases

Canton	**Election data: description and source**	**HB allocation and vote-share calculation: Legal bases**
Aargau	*Grossratswahlen: Detailresultate, ab 1985*: official district results for 2009, 2012, 2016, 2020, and 2024 (Statistisches Amt Kanton Aargau, n.d.). *Detailergebnisse der Grossratswahlen vom 8. März 2009* and *Jahresbericht mit Jahresrechnung 2012*: validation of cantonal seat totals (Departement für Finanzen und Ressourcen des Kantons Aargau, 2009); (Regierungsrat des Kantons Aargau, 2013). *Gemeinsame Grossrats- und Regierungsratswahlen: Vorverfahren gestartet* (2012) and *Ausschreibung der Grossrats- und Regierungsratswahlen 2016*: district seat magnitudes (Staatskanzlei des Kantons Aargau, 2012; 2016a).	*Grossratswahlgesetz (SAR 152.100), version from 1 November 2004*: initial allocation and allocation of remaining seats (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)). *Grossratswahlgesetz (SAR 152.100), Stand 1 January 2009*: rounded district-level voter equivalents for the 2009 election (Kanton Aargau, 1988/2009, § 14(1)). *Grossratswahlgesetz (SAR 152.100), version in force from 1 May 2012* : unrounded district-level voter equivalents from the 2012 election on (Kanton Aargau, 1988/2012, § 14(1)).

Canton	Election data: description and source	HB allocation and vote-share calculation: Legal bases
Graubünden	*Pressedaten Proporzwahl kantonal: Grossratswahlen*: official election-result Excel workbooks for 2022 and 2026 (Kanton Graubünden, 2022; 2026).	*Grossratswahlgesetz (SAR 152.100), version in force from 1 November 2004*: basis for the hypothetical HB allocation (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)). *Verfassung des Kantons Graubünden (BR 110.100), version of 1 January 2018*: explaining the former majority system (Kanton Graubünden, 2003/2018, Art. 27).
Nidwalden	*Wahlergebnis der Landratswahlen*: unpublished Excel workbooks for 2014, 2018, 2022, and 2026, supplied by the Kanzleisekretariat via email (Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026).	*BGE 136 I 352*: Federal Supreme Court description of the former district-level HB allocation (BGE 136 I 352, E. 3.2).
Schaffhausen	*Frühere Urnengänge (seit 2003)*: official result PDFs for 2008, 2012, and 2016 (Kanton Schaffhausen, n.d.). *Resultate der Kantonsratswahl*: official result PDFs for 2020 and 2024 (Staatskanzlei Kanton Schaffhausen, 2020; 2024).	*Proporzwahlverordnung vom 13. November 1979 (Amtsdruckschrift 2734)*: single-seat district rule, HB distribution number, and tie-breaking provisions. Original text supplied by the Staatsarchiv via email (Regierungsrat des Kantons Schaffhausen, 1979, §§ 2, 36–37).
Schwyz	*Kantonsratswahlen 2016: Gemeindeergebnisse*: 30 unpublished municipality-level CSV files supplied via email (Staatskanzlei Kanton Schwyz, 2016). *Kandidierendenresultate: Kantonsratswahlen 2020*: candidate-result CSV export (Staatskanzlei Kanton Schwyz, 2020). *Kantonsprotokoll der Erneuerungswahl Kantonsrat 2024*: election-result Excel workbook (Staatskanzlei Kanton Schwyz, 2024).	*Urteil des Bundesgerichts vom 19. März 2012 (1C_407/2011, 1C_445/2011 und 1C_447/2011)*: description of the former municipality-level HB allocation (Bundesgericht, 19. März 2012, E. 5.2).
Uri	*AuswertungKapitel2-2024*: unpublished Excel workbook containing the 2020 and 2024 results, supplied via email (Standeskanzlei Uri, 2024a). *Landratswahlen 2024: Statistische Auswertung*: official report used for party-label validation (Standeskanzlei Uri, 2024b).	*Synoptische Darstellung zum Gesetz über die Verhältniswahl des Landrats (Proporzgesetz)*: former allocation and tie-breaking rules in the column "Bisher" (Kanton Uri, 2018, Arts. 22–23).
Valais	Official district-level Grand Council result workbooks for 2017, 2021, and 2025 (Excel files). *Grossräte 2017 – Resultate: Offizielle Endresultate*: validation of the 2017 results (Staat Wallis, 2017). Grand Council validation-commission reports: Bericht zur konstituierenden Session vom 19. April 2021 and Rapport relatif à la session constitutive du 14 avril 2025; validation of cantonal vote and seat totals. *Verteilung der Anzahl Grossratssitze pro Bezirk* (legislatures 2017–2021 and 2021–2025) and *Anhang zur Verteilung der Anzahl Grossratssitze pro Bezirk, Legislaturperiode 2025–2029*: district seat magnitudes (Kanton Wallis, 2016; 2020; 2024).	*Gesetz über die politischen Rechte (kGPR; SGS 160.1), version of 9 April 2017*: statutory eligibility threshold and district-level HB allocation (Kanton Wallis, 2004a, Arts. 154–156).

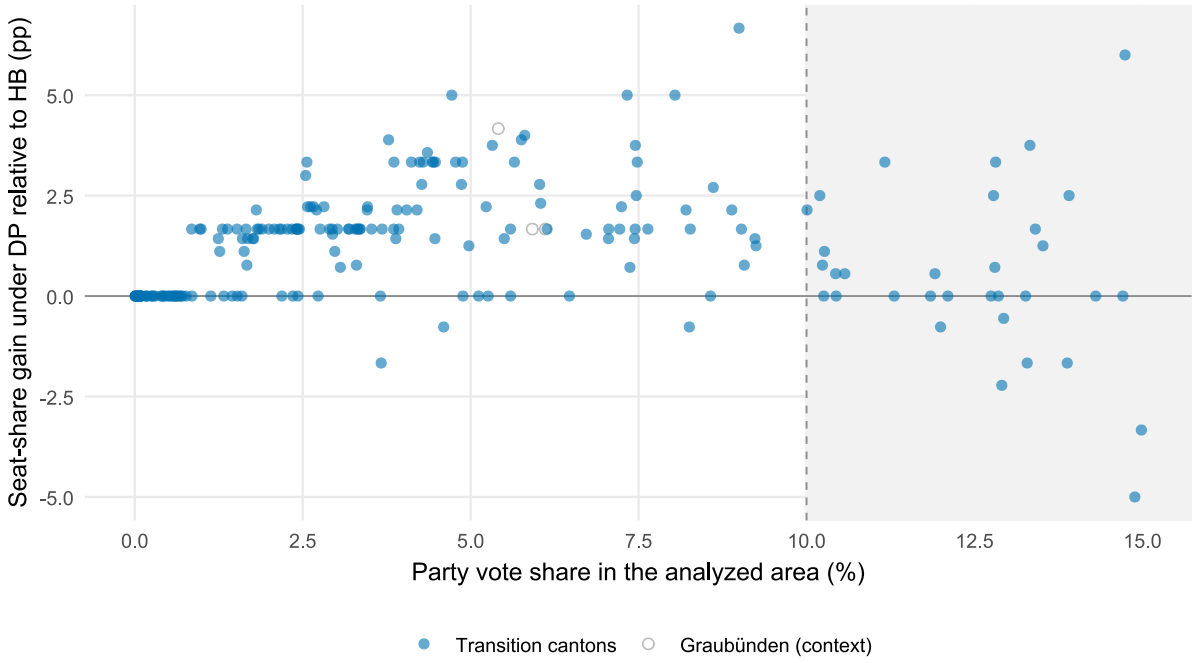
Canton	Election data: description and source	HB allocation and vote-share calculation: Legal bases
Zug	*Kantonsratswahl Kanton Zug* (2014), *Kantonsratswahl 2018*, *Kantonsratswahl 2022*, and *Erneuerungswahl für den Kantonsrat 2026*: official open-data candidate-result CSV exports (Kanton Zug, 2022b; 2022a; 2022c; 2026).	*Wahl- und Abstimmungsgesetz (WAG), original text of 28 September 2006*: former HB and Majority provisions (Kanton Zug, 2006, §§ 46–47, 55–56). *BGE 136 I 376*: confirms Neuheim's former majority system; Neuheim is included using a hypothetical HB allocation (BGE 136 I 376, E. 5.1). *Wahl- und Abstimmungsgesetz (WAG; BGS 131.1), Stand 1 January 2023*: rounded constituency-level voter equivalents for the 2014, 2018, and 2022 elections (Kanton Zug, 2006/2023, § 52d(1)). *Wahl- und Abstimmungsgesetz (WAG; BGS 131.1), Stand 1 January 2026*: unrounded constituency-level voter equivalents for the 2026 election (Kanton Zug, 2006/2026, § 52d(1)).
Zürich	*Archiv Wahlresultate Kantonsratswahlen*: official party-result and candidate-result Excel workbooks for 2007, 2011, 2015, 2019, and 2023 (Amt für Statistik und Daten Kanton Zürich, 2026).	*Gesetz über die Wahlen und Abstimmungen (Wahlgesetz; LS 161), version of 1 January 2005*: former HB allocation and tie-breaking provisions (Kanton Zürich, 1983/2005, §§ 87–88b). *Gesetz über die politischen Rechte (GPR; LS 161), Stand 1 April 2005*: rounded constituency-level voter equivalents for the 2007, 2011, 2015, and 2019 elections (Kanton Zürich, 2003/2005, § 103(1)). *Gesetz über die politischen Rechte (GPR; LS 161), amendment effective 1 October 2021*: unrounded constituency-level voter equivalents for the 2023 election (Kanton Zürich, 2003/2021, § 103(1)).

8.2 Appendix 2: Additional Analysis

8.2.1 Mechanical seat-share gains by exact party vote share

Figure 6 supplements the Figure 2 in the Analysis of the main hypothesis. Conversely to Figure 2 it uses no grouping and plots each party-election observation at its exact vote share. It shows substantial variation in mechanical seat-share gains among parties with similar vote shares as well as no clear monotonic relationship. Additionally to Figure 2 it shows that lists with very low vote shares cluster around zero. Figure 6 therefore provides additional context for the group averages reported in the main analysis. As in the grouped figure, grey open circles show Graubünden's hypothetical HB comparison for context only and the observations do not enter the transition summaries.

Figure 6: Mechanical party-level seat-share gains by exact vote share



Notes: Each point represents one party-election observation with a vote share below 15%. Grey shading marks vote shares outside the smaller-party definition (10% or above).

8.3 Appendix 3: Leave-one-canton-out robustness checks

The leave-one-canton-out check is a sensitivity and robustness check, where the statistical results of the thesis are recalculated after the omission of each single canton. Through that it can be observed whether the results are sensitive to the inclusion of a specific canton or not. The bold values in the following tables identify the canton omission that produces the largest absolute change from the full-sample result in each outcome column. For EQ3, this comparison is made separately within each election number. The comparisons use unrounded results.

8.3.1 Main Hypothesis

Table 5: Leave-one-canton-out robustness: overall smaller-party seat-share gains

Canton omitted	Elections remaining	Mean gain (pp)	Weighted mean gain (pp)
Full sample	31	7.57	7.82
AG	26	7.56	7.88
NW	27	7.70	7.91
SH	26	**6.01**	7.01
SZ	28	7.95	8.21
UR	29	7.99	7.97
VS	28	8.16	8.61
ZG	27	8.22	8.33
ZH	26	6.78	**6.33**

Both the unweighted and weighted mean gains are relatively consistent after the exclusion of every canton. The means are sensitive to the omission of Schaffhausen and Zurich for the unweighted respectively the weighted mean. In both cases the gain declines when they are excluded. This is aligned with the observation that both Zurich and Schaffhausen receive relatively high seat-share gains. Further both of them are included in the observation with the maximum possible five elections. The canton of Zurich further has the largest parliament of all analyzed cantons, giving it extra influence in the weighted mean analysis. The positive aggregate finding reported in Section 4.5.1 therefore remains robust to the exclusion of any single canton.

8.3.2 Exploratory Question 1

Table 6: Leave-one-canton-out robustness: district-structure comparisons

Canton omitted	Mean size	Share below 9
Full sample	−3.06	−1.70
AG	−3.09	−2.34
NW	−4.10	−1.94
SH	**-1.06**	**1.06**
SZ	−2.77	−2.83
UR	−4.64	−0.27
VS	−4.47	−0.50
ZG	−2.33	−3.12
ZH	−2.06	−3.71

Notes: Group differences are calculated as below-mean minus at-or-above-mean gains, in percentage points. The original full-sample cutoffs are retained.

Omitting Schaffhausen produces the largest absolute change in both comparisons. The difference for mean district magnitude remains negative, whereas the difference for the share of districts below nine seats changes from negative to positive. The direction of the latter comparison therefore depends on Schaffhausen's inclusion. These robustness checks demonstrate sensitivity to the sample composition.

8.3.3 Exploratory Question 2

Table 7: Leave-one-canton-out robustness: electoral-quorum comparison

Canton omitted	Without threshold (pp)	With threshold (pp)	Difference (pp)
Full sample	9.75	6.19	3.56
AG	9.79	5.93	3.86
NW	11.29	NA	5.10
SH	**5.52**	NA	**-0.67**
SZ	NA	6.60	3.15
UR	11.43	NA	5.24
VS	NA	6.96	2.79
ZG	NA	7.00	2.75
ZH	NA	**4.23**	5.52

Notes: NA marks a group in which the omitted canton has no elections. Aargau contributes to both groups and is omitted from both. The difference is calculated as without minus with quorum, using the full-sample mean for an unaffected group. Values are in percentage points.

As in the prior robustness checks the cantons of Zurich and Schaffhausen are responsible for the highest absolute differences. The omission of Schaffhausen reverses the group difference. This means that without the Schaffhausen elections, the average seat-gain share is higher for elections with thresholds compared to the ones without. The sample for the without quorum group consists of twelve elections, from which five are from Schaffhausen. Further all Schaffhausen elections experience an above average seat-share gain. Thus is the observed difference sensitive to Schaffhausen's inclusion and should not be interpreted as a stable relationship between an explicit electoral quorum and the smaller party seat-share gains.

8.3.4 Exploratory Question 3

Table 8: Leave-one-canton-out robustness: successive-election summaries

Election number	Canton omitted	Cantons	Mean gain (pp)	Mean change (pp)
1	Full sample	8	5.17	NA
1	AG	7	4.58	NA
1	NW	7	5.43	NA
1	SH	7	4.47	NA
1	SZ	7	5.47	NA
1	UR	7	5.90	NA
1	VS	7	5.68	NA
1	ZG	7	5.55	NA
1	ZH	7	**4.24**	NA
2	Full sample	8	6.80	1.63

Election number	Canton omitted	Cantons	Mean gain (pp)	Mean change (pp)
2	AG	7	6.44	1.87
2	NW	7	7.06	1.63
2	SH	7	**5.39**	**0.92**
2	SZ	7	7.20	1.72
2	UR	7	7.38	1.48
2	VS	7	7.66	1.98
2	ZG	7	7.24	1.69
2	ZH	7	6.02	1.79
3	Full sample	7	8.36	0.97
3	AG	6	8.56	**1.49**
3	NW	6	8.64	0.85
3	SH	6	**6.69**	0.85
3	SZ	6	8.91	0.97
3	VS	6	9.11	0.62
3	ZG	6	9.33	1.34
3	ZH	6	7.25	0.67
4	Full sample	5	9.42	−0.51
4	AG	4	10.52	−0.10
4	NW	4	8.85	**-1.89**
4	SH	4	**7.60**	−0.22
4	ZG	4	10.83	−0.95
4	ZH	4	9.27	0.61
5	Full sample	3	11.08	0.53
5	AG	2	13.06	**-0.28**
5	SH	2	**8.29**	0.79
5	ZH	2	11.90	1.07

Notes: Changes are calculated within cantons before averaging. Only cantons observed at the respective election number are omitted. The full-sample reference row for each election number is shaded light grey. Gains and changes are in percentage points.

For exploratory question three, the robustness check was done for each election number separately. Zurich or Schaffhausen produces the largest absolute change in six out of nine comparisons. The mean seat-share gains remain positive under every omission. However, the mean change from election three to four switches from negative to positive when Zurich is excluded, while the change from election four to five switches from positive to negative when Aargau is excluded. These later transitions include

only five respectively three cantons. This makes their averages particularly sensitive to individual cantons.

8.4 Appendix 4: Declaration of Aids and Assistance

Table 9: Tools and assistance used in the thesis

Resource	Usage	Affected sections
ChatGPT/Codex (mainly 5/6/6.1 Sol)	Used for structuring, advising and coding.	Throughout the thesis and code
DeepL Free	Translating single thoughts and words from German to English.	Throughout the thesis
Claude Free	Used to challenge statements from ChatGPT/Codex	Throughout the thesis
Github CoPilot Free	Used to help with inline comments	Throughout the code
Private proofreading: Martin Beyn, Clara Burmeister, Eric Oehler	Corrections and suggestions concerning language, clarity and readability.	Throughout the thesis

8.4.1 Self Reflection on AI-Usage

Generally, I see AI as a useful tool. It helped me throughout the process of writing this Thesis. As with any technology are there downsides and upsides to the usage of it. On the positive side, I want to highlight that AI helps me to structure my work better. This helps me especially, since it is something where I still can improve. Further, it helped me find discrepancies throughout the code and text, and it does this with a much higher precision than I could. Moreover, the usage of AI models is helpful for recreating repetitive tasks. For example, I used it to help me build the data cleaning files. This was especially helpful, since before writing this thesis, I had never orchestrated and coded such a comprehensive project. Codex, in that case, helped me to build a helper function. Without the usage of an AI, it would have probably taken me longer to come up with the idea of using a helper, respectively to find out about them. Thus, the AI helped me to to build the cleaning and standardizing process quicker and with more precision.

One point of criticism I have when using an AI is that when one challenges a statement of it, the AI often immediately agrees with the user's opinion. For me this phenomenon weakens the trust into the statements of an LLM and makes it harder to reason whether my proposal or the AI's is closer to what's needed. Moreover, I perceive that AI models, when asked to propose a rewording of a paragraph or similar, tend to lenghten the text quite massively. In my perception, the longer text is only providing a limited amount of further value, especially since i saw a tendency towards duplication. The latter phenomenon may come from that the AI possibly only reads the proposed part and has no further knowledge about other relevant parts of the text. In this case of duplicates it is the users (my) obligation to prompt better. Not only when asked to propose improvements to the text did I perceive that it was often longer than needed, but also when asking the AI to help me find weaknesses in the code. In the case of coding, I felt that the AI sometimes wanted to implement fail-safes which, when looking at the scope of my thesis, made no or only little sense. This, again, might be connected to the AI of analyzing only a limited context of the thesis.

To sum up, I see AI as a very helpful tool for the thesis, especially when it comes to outlining both the thesis structure and the code. For me, the main weakness is the as perceived limited reasoning about whether the addition of text/code adds real value or not.

8.5 Appendix 5: Declaration of Authorship

"I hereby declare,

- that I have written this thesis independently,
- that I have written the thesis using only the aids specified in the index;
- that all parts of the thesis produced with the help of aids have been declared;
- that I have handled both input and output responsibly when using AI. I confirm that I have therefore only read in public data or data released with consent and that I have checked, declared and comprehensibly referenced all results and/or other forms of AI assistance in the required form and that I am aware that I am responsible if incorrect content, violations of data protection law, copyright law or scientific misconduct (e.g. plagiarism) have also occurred unintentionally;
- that I have mentioned all sources used and cited them correctly according to established academic citation rules;
- that I have acquired all immaterial rights to any materials I may have used, such as images or graphics, or that these materials were created by me;
- that the topic, the thesis or parts of it have not already been the object of any work or examination of another course, unless this has been expressly agreed with the faculty member in advance and is stated as such in the thesis;
- that I am aware of the legal provisions regarding the publication and dissemination of parts or the entire thesis and that I comply with them accordingly;
- that I am aware that my thesis can be electronically checked for plagiarism and for third-party authorship of human or technical origin and that I hereby grant the University of St.Gallen the copyright according to the Examination Regulations as far as it is necessary for the administrative actions;
- that I am aware that the University will prosecute a violation of this Declaration of Authorship and that disciplinary as well as criminal consequences may result, which may lead to expulsion from the University or to the withdrawal of my title."

By submitting this thesis, I confirm through my conclusive action that I am submitting the Declaration of Authorship, that I have read and understood it, and that it is true.



Moritz Beyn, 11.11.2026

Place and date

Jan Moritz Beyn